

VOICENET: Fine-Grained Voice Understanding Beyond Emotion at Scale

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Abstract

Expressive speech synthesis has outpaced expressive speech perception: systems now render fine-grained vocal performances that no public benchmark can score. Most benchmarks for this inverse problem stop at six to nine basic emotion categories, largely on acted speech. This paper introduces VOICENET, a human-annotated representation-level benchmark for voice performance understanding on permissively-licensed in-the-wild speech. VOICENET has two subsets: VOICENET-EMO applies a 40-emotion taxonomy with three expert ratings per item, and VOICENET-EXT, a preliminary subset, scores 57 talking-style attributes including speaking rate, vocal tension, breathiness, and register. The paper also releases EMOLIA, an emotion-annotated version of the Emilia corpus, with a curated rebalanced subset enriched by dense MOSS-Audio Thinking annotations. Two voice-text contrastive baselines train on this data: a 110M-parameter VOICECLAP-SMALL for fast large-scale data filtering and a 7B VOICECLAP-LARGE for state-of-the-art performance. Both outperform existing CLAP baselines, which sit near chance on VOICENET-EMO. On VOICENET-EMO, VOICECLAP-LARGE aligns more closely with the aggregate expert consensus than individual experts agree with one another: a comparison against the majority label rather than evidence of surpassing human emotion perception. All systems evaluated here are voice-text embedding models: VOICENET scores representation-level attribute recognition and retrieval, not end-to-end spoken-dialogue behaviour. Clustering and filtering uncurated speech corpora into subsets that span diverse talking styles and emotions remains an open challenge; VOICECLAP embeddings offer a promising tool for this task. VOICENET, EMOLIA, and VOICECLAP are publicly available for research use.

1 Introduction

Synthetic speech technology generates fluent, expressive speech (OpenAI, 2024; Kirk et al., 2025), but recognizing fine-grained vocal performance has not kept pace (Cowie et al., 2001; Schuller, 2018); two evaluation gaps slow progress. First, public speech-emotion benchmarks rely on a small set of basic emotions (Ekman, 1992; George & Ilyas, 2024): IEMOCAP (Busso et al., 2008), RAVDESS (Livingstone & Russo, 2018), CREMA-D (Cao et al., 2014), SAVEE (Jackson & Haq, 2014), and EMODB (Burkhardt et al., 2005) cover 6 to 9 emotions on studio-acted speech, and aggregation efforts (Scheidwasser-Clow et al., 2022; Ma et al., 2024; Osman et al., 2024) inherit those limits.

Second, no public benchmark scores talking-style attributes (speaking rate, vocal tension, breathiness, register, chunking, timbre, recording context) alongside emotion, despite their established role in paralinguistic perception (Schuller et al., 2013; Cowie et al., 2001). Synthetic resources such as EMONET-VOICE (Schuhmann et al., 2025a) address taxonomy granularity but use text-to-speech audio; whether models trained on dense annotations transfer to in-the-wild speech is open.

Three observations motivate this paper. The 40-emotion taxonomy grounded in modern affective science (Barrett, 2017; Russell, 1980; Cowen et al., 2019) has been validated on synthetic EMONET-VOICE audio and applies to permissively-licensed real-human speech. Existing CLAP (Wu et al., 2023; Elizalde et al., 2024) and voice-text models (Dinkel et al., 2025; Yang et al., 2026b) are trained on general-audio captions, not dense vocal-style descriptions. Open-source audio language models such as MOSS-AUDIO-8B-THINKING (Yang et al., 2026a) produce structured per-clip annotations across emotion, voice quality, and acoustic context at scale.

This paper contributes: (1) **VOICENET benchmark**. A human-annotated benchmark on permissively-licensed in-the-wild speech with two subsets: VOICENET-EMO extends the EMONET-VOICE 40-emotion taxonomy to 7,988 (audio, prompt) pairs over 3,944 clips with three expert raters per item; VOICENET-EXT covers 57 talking-style attributes¹ over 18.5k (audio, level) pairs carrying 40,990 ratings from eight annotators. (2) **EMOLIA, EMOLIA-BALANCED, and the MOSS-Audio annotation suite**. A fully emotion-annotated version of Emilia (71.78M clips with 40 emotion scores, captions, and speaker embeddings), a curated 5.26M-clip subset (EMOLIA-BALANCED) rebalanced across 40 emotions and 3,000 speaker clusters, and MOSS-AUDIO-8B-THINKING annotations on EMOLIA-BALANCED plus three open voice corpora (8.64M clips, 18 prompt-groups, 61 attribute values per clip). (3) **VOICECLAP models**. VOICECLAP-SMALL is a 110M-parameter dual-tower BUD-E-Whisper-Small + all-MiniLM-L6-v2 trained with SigLIP. VOICECLAP-LARGE is a rank-16 LoRA finetune of LCO-EMBEDDING-OMNI-7B (Xiao et al., 2025) trained with InfoNCE. VOICECLAP-LARGE reaches 0.702 per-prompt balanced accuracy on VOICENET-EMO (+0.039 over the zero-shot Omni base). (4) **Evaluation**. Seven general-audio CLAPs (Wu et al., 2023; Elizalde et al., 2024; Dinkel et al., 2025; Yang et al., 2026b; Niizumi et al., 2025; Li et al., 2024; Zhu et al., 2024) sit at chance on VOICENET-EMO ($|\rho| < 0.1$); two zero-shot Omni-Embedding bases (Xiao et al., 2025) reach 0.65 to 0.66.

Scope. VOICENET is a *representation-level* benchmark: it measures how well voice-text embedding models rank fine-grained emotion and talking-style descriptions against real speech. Every system evaluated in §4 is an audio-text embedding model; end-to-end voice assistants, conversational context, response generation, and generative audio-language models (which require a different scoring interface than cosine similarity) are out of scope.

Beyond the headline benchmark, clustering and filtering raw uncurated speech corpora into subsets that span diverse talking styles and emotions remains an open challenge. VOICECLAP embeddings provide a practical handle on this problem: nearest-neighbour search on the embedding space surfaces clips by emotion, talking-style, and recording context. An interactive browser of EMOLIA-BALANCED² demonstrates this use case.

2 Related Work

Speech emotion benchmarks. Public SER datasets cluster around six to nine acted emotion classes: IEMOCAP (12h, 9 emotions) (Busso et al., 2008), RAVDESS (1h, 8) (Livingstone & Russo, 2018), SAVEE (0.8h, 7) (Jackson & Haq, 2014), EMODB (German, 7) (Burkhardt et al., 2005), and CREMA-D (6) (Cao et al., 2014). Their taxonomies derive from basic-emotion theory (Ekman, 1992; Plutchik, 2001), whereas vocal-expression studies map a far larger space of distinct emotions (Cowan et al., 2019); acted prosody overstates cues (Wilting et al., 2006), and ethical barriers prevent

¹<https://projects.laion.ai/emolia-bench/taxonomy/>

²<https://projects.laion.ai/emolia-bench/demo/>

collection of stigmatising emotions (Schuller et al., 2013). Aggregations such as SERAB (9 corpora, 6 languages) (Scheidwasser-Clow et al., 2022), EMOBOX (32 datasets, 14 languages) (Ma et al., 2024), and SER EVALS (18 minority-language corpora) (Osman et al., 2024) broaden coverage but inherit acted speech and narrow taxonomies; BERST adds shouted speech from 98 actors (Tuttösí et al., 2026), and MSP-PODCAST (Lotfian & Busso, 2019) is one of few in-the-wild corpora at scale. The EMONET-VOICE bench (Schuhmann et al., 2025a) introduces the 40-emotion taxonomy on synthetic audio; VOICENET ports it to in-the-wild human speech, adds 57 talking-style attributes, and uses contrastive cosine-similarity scoring rather than ordinal regression.

Voice-text contrastive models. Contrastive Language-Audio Pretraining was first applied to general-audio sound events (Wu et al., 2023; Elizalde et al., 2024). Subsequent models target singing voice (Yang et al., 2026b) or unify speech, music, and sound (Dinkel et al., 2025). Training data for these models comes from AudioCaps and Clotho, captioned at sound-event level. None target dense vocal-style annotation. §4 confirms that all general-audio CLAPs sit at chance on VOICENET-EMO. Single-tower audio-language embedding models (LCO-EMBEDDING-OMNI-7B (Xiao et al., 2025), LCO-EMBEDDING-OMNI-3B (Xiao et al., 2025)) inherit speech understanding from instruction-tuned multimodal LLMs, reaching approx. 0.65 on VOICENET-EMO zero-shot.

Annotation taxonomies and dimensional models. Affective science models emotions as context-dependent and graded (Barrett, 2017; Lindquist, 2013). Valence-arousal-dominance (Russell, 1980) and multi-label schemes (Zhang et al., 2020; Cowen et al., 2019) support blended affect. Most benchmarks still assign a single discrete label per clip. Intensity annotations show low crowd-source agreement (Kajiwaru et al., 2021; Stappen et al., 2021). VOICENET-EXT takes a complementary route by annotating 57 talking-style attributes with binary level rubrics, broadening the supervision signal beyond emotion categories.

3 VOICENET Suite

The suite has four released components: the 40-emotion + 57-attribute taxonomy (§3.1); EMOLIA (the fully annotated Emilia corpus) with EMOLIA-BALANCED and three sibling MOSS-Audio-annotated training corpora (§3.2); the VOICENET human-annotated benchmark with two subsets at different maturity levels the near-complete, expert-rated VOICENET-EMO and the preliminary VOICENET-EXT, whose inter-rater agreement is at chance (§3.3); and the two VOICECLAP contrastive models (§3.4).

3.1 Taxonomies

Emotion taxonomy (40 categories). The 40-category emotion taxonomy originally developed for EMONET-FACE (Schuhmann et al., 2025b) covers positive emotions (*Elation*, *Contentment*, *Affection*, and *Awe*), negative emotions (*Distress*, *Sadness*, *Bitterness*, and *Contempt*), cognitive states (*Concentration*, *Confusion*, and *Doubt*), physical states (*Pain*, *Fatigue*), and socially mediated emotions (*Embarrassment*, *Shame*, *Pride*, and *Teasing*). The full set of 40 categories with descriptive terms appears in App. A.3. The literature-extraction and expert-guided refinement process is detailed in App. A.11.

Talking-style attribute taxonomy (57 attributes). VOICENET-EXT covers a parallel 57-attribute taxonomy of talking-style and paralinguistic descriptors, grouped into perceived speaker traits, affective dimensions, prosodic delivery, vocal quality, resonance placement, recording context, and style descriptors (full list with short codes in App. A.4). The 57 attributes derive from the 61-value MOSS-AUDIO schema (§3.2) by dropping vocal-burst presence (BURST, redundant with discrete event detection), fine-grained categorical emotion (EMO, covered by VOICENET-EMO), accent (ACNT, demographic rather than perceptual), and language (LANG, categorical rather than ordinal). The rubric-level definitions used for human annotation are available at <https://>

3.2 EMOLIA, EMOLIA-BALANCED, and the MOSS-AUDIO Annotation Pipeline

EMOLIA is a fully emotion-annotated version of the open-source Emilia corpus. The complete Emilia-Large release (71.78M clips, 215,600 hours; its Emilia portion is licensed CC-BY-NC-4.0 and its Emilia-YODAS portion CC-BY-4.0) is annotated with 40 emotion scores from the EmpathicInsight-Voice classifier, emotion captions from BUD-E-Whisper, and WavLM-based speaker-timbre embeddings. EMOLIA is released in full as a large-scale training resource for emotion-aware speech research.

EMOLIA-BALANCED is a curated 5.26M-clip subset of EMOLIA, rebalanced across the 40 emotions and 3,000 k-means speaker-embedding clusters using empathic-voice-classifier logits. Together with 3 additional open voice corpora (LAION’s Got Talent 1.69M clips, Majestrino 973k clips, Multilingual In The Wild 721k clips), EMOLIA-BALANCED forms the MOSS-Audio annotation suite (8.64M clips total; per-corpus counts in App. A.2, Tab. 4). An internal ablation excluded Multilingual In The Wild from the training mixture (7.92M clips) without dropping it from the release.

Annotation pipeline. Each clip is queried with 18 prompt-groups covering related attribute clusters (resonance placement, prosodic delivery, recording context, style category, etc.). MOSS-AUDIO-8B-THINKING (Yang et al., 2026a) returns a structured response per group; the parser consumes only the output after the closing `</think>` tag and regenerates malformed groups. The 18 outputs supply 61 short-code attribute values per clip, concatenated into one `text` field for contrastive training. The chain-of-thought traces emitted before `</think>` are retained and released alongside the structured annotations (155.46M traces total) to support future work on reasoning-supervised voice understanding; they are not used for the contrastive training in this paper.

EMOLIA-BALANCED construction. The Emilia source corpus is filtered with *EmpathicVoice* released with EMONET-VOICE bench (Schuhmann et al., 2025a), a Whisper-based audio encoder with a 40-dim. linear head trained on synthetic EMONET-VOICE-Voice to predict per-clip presence logits. The classifier is used for stratified sampling and benchmark prompt generation, never as ground truth. Clips are sampled to balance the 40-emotion categories and 3,000 k-means clusters of WavLM speaker embeddings. Rebalancing softens the source-corpus emotion skew of Emilia but does not fully remove it.

Training corpora. VOICECLAP trains on nine voice corpora: EMOLIA-BALANCED (5.26M), LAION’s Got Talent (1.69M), Majestrino (973k), two in-house Synthetic Vocal Bursts collections (341k), and four FCaps-captioned corpora (Yang et al., 2026b): EARS (17k), Espresso (27k), Voxceleb1 (154k), Voxceleb2 (600k). The smaller six broaden the contrastive negative distribution; their captions are not part of the EMOLIA suite. Full counts in App. A.5, Tab. 5.

3.3 VOICENET: A Human-Annotated In-the-Wild Benchmark

VOICENET evaluates fine-grained voice understanding on permissively-licensed in-the-wild human speech. Unlike the prior EMONET-VOICE bench (Schuhmann et al., 2025a), which evaluates on synthetic TTS-generated audio, VOICENET consists exclusively of real human vocal performances; the VOICENET-EMO clips come from the Emilia-YODAS (CC-BY-4.0) and Emilia (CC-BY-NC-4.0) portions of Emilia-Large, so the benchmark is openly available under CC-BY and CC-BY-NC research licences. Each clip is paired with a textual prompt. Raters mark whether the property described by the prompt is present in the clip. Two subsets share this protocol but differ in prompt vocabulary, audio source, and rater pool.

VOICENET-EMO (40 emotions, expert raters). VOICENET-EMO applies the same 40-emotion

taxonomy as EMONET-VOICE but to naturalistic, in-the-wild human speech rather than synthetic audio. It asks: *is emotion E present in clip C?* for each of the 40 emotions. The subset contains 7,988 (audio, emotion) pairs over 3,944 unique audio clips and 40 emotion prompts. 7,984 of those pairs received the full three independent ratings from psychology experts on the three-point scale used in prior work (Schuhmann et al., 2025a): 0 (not present), 1 (weakly present), and 2 (strongly present); the remaining 4 pairs received one or two ratings due to incomplete rater coverage and are excluded from inter-rater statistics but retained in the released benchmark. Three experts contributed (`user_0`, `user_1`, and `user_2`). Each (audio, emotion) pair was rated independently with assignments balanced for rater gender. Annotators were blinded to one another’s ratings.

Each pair is sampled under one of five task types so that emotion presence is balanced across the subset. *Affirmative* (4,000 pairs) presents an emotion that the empathic voice classifier (§3.2) predicts is present. *Contrastive 1* and *contrastive 2* (1,000 each) present natural opposites of the predicted emotion (e.g., “anger” and “wrath” for a clip predicted as “happiness”). *Ultimate* (995) and *penultimate* (993) present the lowest- and second-lowest-ranked emotions. Experts confirm 78.85% of affirmative pairs by majority vote, and 0.349, 0.372, 0.376, and 0.371 of the contrastive and ultimate/penultimate pairs, consistent with classifier-ranking noise and blended affect; the aggregate majority-present rate is 0.578. Because the classifier is trained on synthetic EMONET-VOICE-Voice audio, preselection could bias which clips and emotions enter the benchmark. The five balanced task types and the use of the independent expert majority vote, never the classifier prediction, as the released label mitigate but do not remove this circularity; the classifier acts as a sampling prior, not as ground truth. Inter-rater agreement (Fleiss’ binary κ) is 0.086, in line with the low end reported for fine-grained intensity annotation (Kajiwar et al., 2021; Stappen et al., 2021); pairwise and leave-one-out human-consistency references are 0.562 and 0.572 balanced accuracy. App. A.12 reports per-emotion agreement.

VOICENET-EXT (57 attributes, eight mixed-background raters; preliminary). VOICENET-EXT asks: *does clip C exhibit attribute A at level L?* for the 57 talking-style attributes of §3.1. The audio source is distinct from EMOLIA-BALANCED. Each attribute is defined by a rubric with ordinal levels (seven levels, 0–6, for nearly all attributes). For each (attribute, level) bucket, Gemini 3 Flash pre-screens candidate clips, and human annotators confirm or reject the match for an equal number of positives (clips matching the target level) and negatives (clips matching a non-adjacent level), so each judgement is binary rather than a full ordinal rating. The release contains 18.5k (audio, attribute-level) pairs carrying 40,990 ratings from eight annotators; four hold psychology degrees and contributed 95.3% of all ratings. 5,828 pairs carry at least three ratings, 10,338 carry two, and 2,366 carry one. The aggregate majority-yes rate is 0.284. On the 5,583 items with exactly three ratings, Fleiss’ binary $\kappa = -0.022$ (95% CI $[-0.037, -0.007]$), an interval that excludes zero, so near-chance agreement is a property of these rubric-level judgements rather than an artefact of sparse annotation. Under a reliable-core rule prespecified before computing per-attribute results (at least 20 three-rater items, $\kappa \geq 0.20$, and a 95% lower bound above zero), no attribute qualifies; the maximum is $\kappa=0.085$ for STNC. All VOICENET-EXT scores are therefore preliminary and its attributes are exploratory perceptual probes. App. A.14 gives the full coverage and reliability analysis.

3.4 VOICECLAP: Voice-Text Contrastive Models

VOICECLAP-SMALL. A 110M-parameter dual-tower CLAP (Wu et al., 2023): a BUD-E-Whisper-Small (Radford et al., 2023) audio encoder (768-d) and an `all-MiniLM-L6-v2` (Reimers & Gurevych, 2019; Wang et al., 2020) text encoder (384-d, mean-pooled), each linearly projected to a shared 768-d space and trained for one epoch with the SigLIP sigmoid contrastive loss (Zhai et al., 2023).

VOICECLAP-LARGE. A rank-16 LoRA (Hu et al., 2022) finetune of the single-tower LCO-EMBEDDING-OMNI-7B (Xiao et al., 2025) (Qwen2.5-Omni-Thinker-7B backbone, 3,584-d output), trained for one epoch with symmetric InfoNCE (van den Oord et al., 2018). Optimiser settings, batch

Table 1: Headline results on VOICENET, grouped by model family and sorted within each group by VOICENET-EMO per-prompt balanced accuracy ($bal@pp$). ρ is the mean per-prompt Spearman correlation between cosine similarity and human present-vote share. All numbers are computed on items with at least two human raters (7,986 of 7,988 VOICENET-EMO items and 16,166 of 18,532 VOICENET-EXT non-LANG items) so the labels reflect a multi-rater majority rather than a single annotator’s call; per-tag files in App. A.7 report both this filtered cut and the full-set numbers. Bootstrap CIs are in Tab. 6. Human baselines and consistency references on the same task are discussed in the text. Best per column in bold; higher is better.

Model	VOICENET-EMO $bal@pp$	VOICENET-EXT $bal@pp$	VOICENET-EMO ρ	VOICENET-EXT ρ
Cacophony (Zhu et al., 2024)	0.5432	0.6089	−0.0183	0.0188
LAION-CLAP (Wu et al., 2023)	0.5624	0.6184	0.0310	0.0600
CLSP (Yang et al., 2026b)	0.5675	0.6221	0.0610	0.0937
M2D-CLAP-2025 (Niizumi et al., 2025)	0.5681	0.6361	0.0348	0.1003
MS-CLAP-2023 (Elizalde et al., 2024)	0.5754	0.6321	−0.0084	0.1044
GLAP (Dinkel et al., 2025)	0.5765	0.6254	0.0667	0.0836
MGA-CLAP (Li et al., 2024)	0.5897	0.6208	0.0947	0.0819
LCO-Embedding-Omni-3B (Xiao et al., 2025)	0.6547	0.6293	0.2876	0.0873
LCO-Embedding-Omni-7B (Xiao et al., 2025)	0.6632	0.6342	0.3052	0.0977
ours VOICECLAP-SMALL	0.6754	0.6367	0.3176	0.1051
ours VOICECLAP-LARGE	0.7021	0.6510	0.3719	0.1475

sizes, and compute for both models are given in App. A.13.

4 Experiments

Evaluation harness. For each model, embeddings are produced for all audio clips and all prompts in a subset. Both are L2-normalised. Pairwise cosine similarity yields a matrix $S \in \mathbb{R}^{|\text{audios}| \times |\text{prompts}|}$. Each row of the bench’s label table is one (audio, prompt) pair. From S this paper reports four metrics. *Balanced accuracy at threshold 0* ($bal@0$) thresholds raw cosine at 0.0, meaningful only for already-calibrated models. *Balanced accuracy at the global optimal threshold* ($bal@opt$) sweeps unique similarities for the single threshold that maximises balanced accuracy. *Per-prompt balanced accuracy* ($bal@per_prompt$) sweeps a separate threshold per prompt (with a 10-row minimum, falling back to the global threshold), removing per-prompt prior mismatch under our balanced sampling protocol. These Table 1 thresholds are fitted and scored on the same labels, so $bal@per_prompt$ is an oracle-calibrated separability measure; we additionally report five-fold clip-grouped held-out calibration below. *Per-prompt mean Spearman ρ* averages, over prompts, ρ between similarity and the present-vote-share among raters; it is threshold-free and our primary unbiased ranking statistic. $Bal@0$ and $bal@opt$ are reported in App. A.7.

Models and baselines. The two finetunes VOICECLAP-SMALL and VOICECLAP-LARGE (§3.4) are compared against two zero-shot Omni-Embedding bases (LCO-EMBEDDING-OMNI-7B and LCO-EMBEDDING-OMNI-3B (Xiao et al., 2025); abbreviated LCO-OMNI-7B/3B in figure labels) and seven general-audio CLAPs: LAION-CLAP (Wu et al., 2023), MS-CLAP-23 (Elizalde et al., 2024), CLSP (Yang et al., 2026b) (SPEAR-XLarge (Yang et al., 2025) + RoBERTa (Liu et al., 2019)), GLAP (Dinkel et al., 2025) (Dasheng (Dinkel et al., 2024) + SONAR (Duquenne et al., 2023)), M2D-CLAP-2025 (Niizumi et al., 2025), MGA-CLAP (Li et al., 2024), and CACOPHONY (Zhu et al., 2024). Loader correctness for the 7 CLAPs is verified against reference implementations (App. A.6).

4.1 Where the Speech-Emotion Gap Lives

The 40-emotion ranking task is out of distribution for general-audio CLAPs (trained on AudioCaps/Clotho sound-event captions) and only partially in distribution for speech-aware Omni-Embedding bases. Three tiers on the same harness localise the gap: general-audio CLAPs (transfer

lower bound), zero-shot Omni-Embedding bases (speech-coverage pretraining alone), and the VOICECLAP finetunes (dense-caption contrastive alignment on top).

Table 1 separates the three tiers on VOICENET-EMO. General-audio CLAPs cluster at chance ($|\rho| \lesssim 0.1$); MS-CLAP and Cacophony post mildly negative ρ , the expected failure mode when a sound-event model ranks speech clips by emotion presence. The Omni bases reach 0.65–0.66 *bal@pp*. VOICECLAP-LARGE tops the table at 0.702 *bal@pp*, +0.039 over its zero-shot base (Xiao et al., 2025). VOICECLAP-SMALL (110M parameters) lands within ~ 0.03 of its 7B sibling and clears every general-audio CLAP, so the headline is not driven by scale. VOICECLAP-LARGE tops every column on both subsets. A 2,000-replicate paired clip bootstrap (Tab. 6) supports each top-vs.-runner-up gap: VOICENET-EMO differences are +0.0267 *bal@pp* (95% CI [0.0167, 0.0363]) and +0.0543 ρ [0.0360, 0.0714]; the preliminary VOICENET-EXT differences are +0.0143 *bal@pp* [0.0040, 0.0244] and +0.0424 ρ [0.0234, 0.0593]. Most smaller adjacent baseline gaps overlap zero.

Human baselines and ceilings. Random and always-predict-majority both give 0.500 balanced accuracy. The pairwise human reference (rater A’s binary vote scored against rater B’s) is 0.562 on VOICENET-EMO and 0.533 on VOICENET-EXT; the leave-one-out (LOO) reference (one rater against the majority of the others) is 0.572 and 0.479. VOICECLAP-LARGE’s 0.7021 *bal@pp* on VOICENET-EMO is above the LOO reference, i.e. closer to the multi-rater consensus than any individual expert is. This is not a like-for-like comparison, since the model is scored against the aggregate majority label while each expert is scored against the other raters; it indicates strong alignment with the expert consensus, not surpassing human emotion perception. On VOICENET-EXT the human references sit at chance (LOO 0.479), so its model scores measure exploratory separability rather than a validated ceiling comparison. Vote counts and raw-accuracy baselines are in App. A.15.

Calibration. Because *bal@pp* fits thresholds on the evaluated labels, we also report five-fold clip-grouped held-out calibration. On VOICENET-EMO, held-out *bal@pp* is 0.6820 for VOICECLAP-LARGE and 0.6412 for its LCO-Omni-7B base (0.7021 and 0.6632 in-sample), a paired held-out lead of +0.0409 (95% CI [0.0296, 0.0518]). On VOICENET-EXT, held-out scores drop to 0.5596 and 0.5397 while the lead remains +0.0199 [0.0095, 0.0308]; this ≈ 0.09 optimism across model families is why current VOICENET-EXT thresholded scores are exploratory, while threshold-free ρ is unaffected. Threshold-0 and global-threshold results are discussed in App. A.15.

4.2 Per-Emotion Performance on VOICENET-EMO

A wider per-emotion breakdown for VOICECLAP-SMALL, VOICECLAP-LARGE, the two Omni-Embedding bases, and all seven general-audio CLAP baselines is reported in App. A.8, Tab. 7. This appendix view keeps the main text focused on aggregate behavior while exposing which emotions drive the VOICENET-EMO gains in Tab. 1. Three patterns are worth flagging here. (i) Acoustically concrete categories (*distress*, *anger*, *impatience*, *sadness*) cross $\rho=0.5$ for at least one model; (ii) VOICECLAP-LARGE wins or runs second on most rows but *regresses* below the LCO bases on *Teasing*, *Interest*, and *Thankfulness/Gratitude*, suggesting that contrastive finetuning on EMOLIA-BALANCED trades off some categories that the base instruction-tuned model handled well via lexical priors; (iii) the lowest-signal categories across all eleven models include *Doubt*, *Sourness*, *Interest*, *Sexual Lust*, and *Jealousy/Envy* (best ρ across the table is ≤ 0.29). *Sourness* is partly definitional — the taxonomy notes its primarily gustatory origin (App. A.3) — but *Doubt* and *Interest* are mainstream affective categories whose paralinguistic correlates remain weak in this benchmark; we view them as open challenges rather than failure modes of any particular model.

4.3 Per-Attribute Performance on VOICENET-EXT

VOICENET-EXT is harder than VOICENET-EMO at every level: mean per-prompt ρ is 0.15 (VOICECLAP-LARGE) and 0.11 (VOICECLAP-SMALL) versus 0.37 / 0.32 on emo. Two factors compound: prompt density is lower (VOICENET-EXT has 18.5k labelled pairs across 399

(attribute, level) prompts $\Rightarrow \sim 46$ pairs per prompt vs. ~ 200 pairs per emotion on VOICENET-EMO), and the human consistency reference is itself at chance (LOO balanced accuracy 0.479 vs. 0.572 for the expert-rated emo subset). The per-attribute reliability analysis finds no attribute whose fair-or-better κ is supported by a 95% interval above zero, and no attribute reaching $\kappa \geq 0.20$ at all; accordingly, these aggregate model numbers describe exploratory separability on the current labels, not a finalized benchmark ranking. Per-attribute model ρ and reliability tables are released alongside the paper; see App. A.9.

Agreement-stratified results. Splitting the emotions and attributes into lower- and higher-agreement halves before inspecting model scores, none of the four paired high-minus-low interactions for the VOICECLAP-LARGE advantage over LCO-EMBEDDING-OMNI-7B excludes zero (App. A.15); these data do not show VoiceCLAP’s edge concentrating in higher-agreement categories.

4.4 Cross-Dataset Evaluation on Public Speech-Emotion Benchmarks

Ten of the eleven models (all except LCO-EMBEDDING-OMNI-3B) are evaluated against four established speech-emotion datasets to test transfer beyond VOICENET. EmoNet-Voice (Schuhmann et al., 2025a) provides 12,600 synthetic clips across 40 fine-grained emotion classes (the synthetic v1 release that motivates VOICENET’s 40-emotion taxonomy). IEMOCAP (Busso et al., 2008) contains 10,039 dyadic acted utterances over nine emotion labels. RAVDESS (Livingstone & Russo, 2018) contains 1,440 acted speech clips covering eight emotions across twenty-four professional actors. CREMA-D (Cao et al., 2014) contains 7,442 acted utterances from ninety-one actors over six emotions. Each clip is encoded with the model’s audio tower, each label is encoded with an emotion-prompt template, and the prediction is the argmax of cosine similarity. Top-1 accuracy is reported in Table 2.

On EmoNet-Voice all seven general-audio CLAPs sit at chance ($0.024\text{--}0.036$, $1/40 = 0.025$), confirming that the 40-class fine-grained taxonomy is out of distribution for sound-event captioning pretraining. The two voice-aware models lift sharply: LCO-EMBEDDING-OMNI-7B reaches 0.167 (best), VOICECLAP-LARGE reaches 0.155 (runner-up), and VOICECLAP-SMALL reaches 0.105—each $\sim 3.5\text{--}7\times$ above the CLAP baselines. VOICECLAP-LARGE ranks first on IEMOCAP Emotion at 0.321 and on CREMA-D Emotion at 0.511, with the CREMA-D margin to the next-best general-audio CLAP (LAION-CLAP at 0.366) reaching 0.145. On RAVDESS Emotion VOICECLAP-LARGE reaches 0.296, second behind LCO-EMBEDDING-OMNI-7B (Xiao et al., 2025) (0.319). The seven general-audio CLAPs cluster between 0.105 and 0.259 on RAVDESS and 0.147 to 0.231 on IEMOCAP. The Omni base occupies the top tier on EmoNet-Voice, RAVDESS and IEMOCAP but drops to 0.21 on CREMA-D, where six of the seven general-audio CLAPs trained on sound-event captions reach 0.33 to 0.37. Voice-text contrastive finetuning on EMOLIA-BALANCED closes the CREMA-D gap left by the Omni base ($0.209 \rightarrow 0.511$ from LCO-EMBEDDING-OMNI-7B to VOICECLAP-LARGE) while keeping IEMOCAP well above the Omni base (0.321 vs. 0.259). VOICECLAP-LARGE therefore wins two of four external benchmarks and remains within 0.05 of the leader on the other two.

Cross-dataset gap diagnosis. Paired follow-up analyses (App. A.15) indicate that VOICECLAP-LARGE’s deficits against its base on EmoNet-Voice and RAVDESS are class-specific boundary shifts rather than a generic acted-versus-naturalistic effect; on RAVDESS, macro F1 rises from 0.285 to 0.330 despite the lower micro accuracy.

5 Discussion

Dense voice-text supervision closes the speech-emotion transfer gap. All seven CLAP baselines sit at $|\rho| < 0.1$ on VOICENET-EMO, including the speech-specialised CLSP (Yang et al., 2026b) ($\rho=0.061$, $0.568\text{ }bal@pp$), which VOICECLAP-LARGE surpasses by $+0.311\text{ } \rho$ and $+0.135\text{ } bal@pp$

Table 2: Zero-shot emotion classification accuracy on four public speech-emotion benchmarks. EmoNet-Voice is the synthetic 40-class v1 release (Schuhmann et al., 2025a). VOICECLAP-LARGE substantially outperforms other models on IEMOCAP and CREMA-D and is the runner-up on EmoNet-Voice and RAVDESS. Best per column in bold, runner-up underlined; higher is better.

Model	EmoNet-Voice	IEMOCAP	RAVDESS	CREMA-D
LAION-CLAP (Wu et al., 2023)	0.028	0.201	0.173	<u>0.366</u>
MS-CLAP-2023 (Elizalde et al., 2024)	0.025	0.217	0.259	0.355
CLSP (Yang et al., 2026b)	0.036	0.231	0.194	0.333
GLAP (Dinkel et al., 2025)	0.025	0.147	0.115	0.198
MGA-CLAP (Li et al., 2024)	0.029	0.183	0.105	0.349
Cacophony (Zhu et al., 2024)	0.024	0.181	0.181	0.342
M2D-CLAP-2025 (Niizumi et al., 2025)	0.030	0.200	0.184	0.336
LCO-Embedding-Omni-7B (Xiao et al., 2025)	0.167	<u>0.259</u>	0.319	0.209
ours VOICECLAP-SMALL	0.105	0.180	0.210	0.323
ours VOICECLAP-LARGE	<u>0.155</u>	0.321	<u>0.296</u>	0.511

and the $60\times$ smaller VOICECLAP-SMALL by $+0.257$ and $+0.108$. Neither sound-event captions nor speech-aware pretraining alone provide dense vocal-style supervision. The Omni-Embedding bases, pretrained on instruction-tuned multimodal data with substantial speech coverage, already reach $0.65\text{--}0.66$ *bal@pp* zero-shot; contrastive finetuning on EMOLIA-BALANCED adds $+0.039$, and VOICECLAP-SMALL reaches 0.6754 with about $60\times$ fewer parameters, so the gain is not an artefact of scale.

Annotation ambiguity bounds model performance. With Fleiss’ binary $\kappa=0.086$ on VOICENET-EMO, low expert agreement reflects the semantic ambiguity of a 40-way taxonomy on naturalistic speech, consistent with prior fine-grained intensity work (Kajiwara et al., 2021; Stappen et al., 2021). Emotions with concrete acoustic signatures (distress, impatience, anger, and sadness) exceed $\rho=0.5$ for at least one model, while interest, doubt, and sexual lust are among the least stable across models; per-emotion expert unanimity does not predict which categories are hard (App. A.12).

Effect sizes in context. In individual-differences research the 25th, 50th, and 75th percentile correlations are $r=.11$, $.19$, and $.29$ (Gignac & Szodorai, 2016). The general-audio CLAPs fall below the 25th percentile, the Omni bases ($\rho=0.29\text{--}0.31$) reach the 75th, and VOICECLAP-LARGE ($\rho=0.372$) exceeds it, with per-emotion peaks of $0.57\text{--}0.63$ for distress, impatience, anger, and sadness. Because perceived emotion has no objective ground truth (Rum & Perry, 2020; Zaki et al., 2009), these values describe a rank-ordered signal extracted from inherently subjective ratings (App. A.16).

Talking-style attribution is harder than emotion attribution. VOICENET-EXT mean per-prompt ρ is 0.15 for VOICECLAP-LARGE versus 0.37 on VOICENET-EMO, reflecting fewer labelled pairs per prompt (~ 46 vs. ~ 200) and chance-level rater consistency (LOO 0.479). Because agreement stays at chance across $5,583$ three-rater items, the bottleneck is the rubric and the perceptual task rather than annotation volume; sharper rubric definitions and anchor-example rater training are the next step. VOICENET-EMO is therefore the mature benchmark contribution and VOICENET-EXT a preliminary extension whose rankings may change.

Limitations. The evaluation is representation-level: all eleven systems are voice-text embedding models, and no spoken-dialogue or generative audio-language system is evaluated. The MOSS-Audio training annotations are model-generated; an interim prespecified human audit finds that a rater matches the MOSS level about as often as two raters match each other (paired difference $+0.018$, 95% CI $[-0.053, +0.102]$; App. A.10), which bounds rather than establishes label quality. VOICENET-EXT agreement remains at chance and none of its attributes passes the reliable-core rule, so its scores should not be used for confirmatory model ranking. The bootstrap CIs condition on the aggregate labels and fitted thresholds, and seed variance is measured for VOICECLAP-SMALL

only. The source corpora have non-uniform geographic, demographic, and recording-environment coverage. App. A.17 discusses these points in full.

6 Conclusion

This paper introduces VOICENET, a human-annotated benchmark on permissively-licensed real human speech that combines a 40-emotion expert-rated subset (VOICENET-EMO) with a 57-attribute talking-style subset (VOICENET-EXT). It also releases EMOLIA, an emotion-annotated version of the Emilia corpus, the curated balanced subset EMOLIA-BALANCED with dense MOSS-Audio annotations, and three sibling annotated voice corpora. VOICECLAP-SMALL (110M, dual-tower SigLIP) and VOICECLAP-LARGE (7B LoRA finetune of LCO-Embedding-Omni-7B (Xiao et al., 2025) with InfoNCE) reach 0.675 and 0.702 VOICENET-EMO per-prompt balanced accuracy, with mean Spearman $\rho=0.318$ and 0.372 . These correlations exceed the 75th percentile of published psychological effect sizes and surpass speech-specialised CLSP by over $+0.31 \rho$, while all seven evaluated CLAP baselines remain near chance. Dense vocal-style supervision delivers emotion- and attribute-ranking capabilities that neither general-audio pretraining nor speech-aware encoders provide on their own. Clustering and filtering uncurated speech corpora into subsets covering diverse talking styles and emotions remains an open challenge; VOICECLAP embeddings offer a promising tool, demonstrated through the interactive EMOLIA-BALANCED browser.

Acknowledgements

We gratefully acknowledge the support of Intel (oneAPI Center of Excellence), DFKI, Nous Research (providing cluster access and compute), TU Darmstadt, TIB–Leibniz Information Centre for Science and Technology, and hessian.AI (providing compute and helpful discussions), and the open-source community for contributing to emotional AI. This work benefited from the ICT-48 Network of AI Research Excellence Center “TAILOR” (EU Horizon 2020, GA No 952215), the Hessian research priority program LOEWE within the project WhiteBox, the HMWK cluster projects “Adaptive Mind” and “Third Wave of AI”, and from the NHR4CES. Furthermore, this work was partly funded by the Federal Ministry of Education and Research (BMBF) project “XEI” (FKZ 01IS24079B).

AI use statement

AI models were used in the following parts of this work. *Synthetic data and labels.* MOSS-AUDIO-8B-THINKING (Yang et al., 2026a) generated the dense per-clip attribute annotations and chain-of-thought traces released with the four MOSS-annotated corpora (§3.2); a text-only pass of the same model mapped these free-text annotations onto rubric levels for the human audit (App. A.10). The 40 emotion scores and emotion captions in EMOLIA were produced by the EmpathicInsight-Voice classifier and BUD-E-Whisper. These labels are model outputs and are never used as benchmark ground truth; their quality is bounded by the human audit in App. A.10. *Research execution.* GPT-4 extracted candidate emotion terms from text segments during the construction of the 40-category taxonomy (App. A.11), which experts then clustered and refined. Gemini 3 Flash pre-screened candidate clips for each (attribute, level) bucket of VOICENET-EXT before human confirmation (§3.3), and the EmpathicInsight-Voice classifier pre-selected candidate prompts for VOICENET-EMO; every released VOICENET label is a human rating. LLM-based coding assistants helped implement evaluation and statistical-analysis scripts; all reported numbers were regenerated from these scripts and cross-checked against the released per-tag files. *Retrieval.* Claude Opus 5.5 was used to check every reference against Crossref and other public bibliographic databases and to locate the published versions of cited preprints; the authors verified each resulting reference against its source. *Writing.* LLMs assisted with grammar, clarity, and rephrasing of author-written text; no sections were drafted by an LLM. All text, claims, and numerical results were checked by the authors, who take full responsibility for the content.

Ethics statement

VOICENET, EMOLIA, EMOLIA-BALANCED, and the three sibling MOSS-annotated corpora consist exclusively of audio that the source projects (Emilia, LAION’s Got Talent, Majestrino, and Multilingual In The Wild) collected and released under open CC-BY or CC-BY-NC research licences; derived releases inherit the licence of their source clips. No personally identifiable speaker metadata is added beyond the speaker-embedding clusters used for EMOLIA-BALANCED balancing. The 40-emotion taxonomy includes sensitive affective categories (intoxication, sexual lust, malevolence, pain, and shame); the dense MOSS-Audio supervision and the preliminary VOICENET-EXT annotations, whose inter-rater agreement remains at chance, should be treated as research artefacts rather than diagnostic instruments. Audio understanding models trained on EMOLIA-BALANCED could enable affective monitoring with privacy implications; the VOICECLAP-SMALL and VOICECLAP-LARGE checkpoints are released under CC-BY-4.0 for research use and should be paired with downstream safeguards (Helff et al., 2025; Hintersdorf et al., 2024) when deployed.

Reproducibility statement

All evaluation code, training scripts, and corpus manifests are available at <https://github.com/LAION-AI/emolia-bench> and <https://github.com/LAION-AI/voicenet>; the former also hosts the VOICENET benchmark labels (VOICENET-EMO, VOICENET-EXT) and the per-tag files used for every reported number. The EmpathicInsight-Voice classifiers that produced the EMOLIA emotion scores are available at <https://huggingface.co/laion/Empathic-Insight-Voice-Small> and <https://huggingface.co/laion/Empathic-Insight-Voice-Plus>, together with an annotation and inference toolkit at <https://github.com/LAION-AI/emotion-annotations>. Predictors for the 57 VOICENET-EXT dimensions are available at <https://huggingface.co/laion/voicenet-dimension-predictors-commercial>, and Gemini-generated VOICENET-EXT dimension annotations for 236,613 EMOLIA clips at <https://huggingface.co/datasets/laion/emolia-voicenet-gemini-annotations>. The full EMOLIA corpus (71.78M clips) is available at <https://huggingface.co/datasets/laion/Emolia> and EMOLIA-BALANCED at

<https://huggingface.co/datasets/laion/emolia-balanced-5M-subset>; the VOICECLAP-SMALL and VOICECLAP-LARGE checkpoints are available at <https://huggingface.co/VoiceNet/voiceclap-small> and <https://huggingface.co/VoiceNet/voiceclap-large>. The MOSS-Audio-8B-Thinking annotations of EMOLIA-BALANCED, including the reasoning traces, are available at <https://huggingface.co/datasets/VoiceNet/emolia-thinking>; those of the three sibling corpora will be released for research use. The VOICENET-EXT taxonomy and an interactive EMOLIA-BALANCED browser are available at the links in §1. Training hyperparameters are given in §3.4, the data mixture in App. A.5, and the evaluation harness in App. A.6.

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A Appendices

A.1 Comparison to Prior Speech-Emotion Benchmarks

Table 3 situates VOICENET-EMO and VOICENET-EXT against prior speech-emotion benchmarks. The table covers benchmarks only; the MOSS-Audio-annotated training corpora (EMOLIA-BALANCED, LAION’s Got Talent, Majestrino, Multilingual In The Wild) are summarised separately in App. A.2.

Table 3: **Comparison of speech-emotion benchmarks.** #Labels count the size of the label set: emotion or affect categories for prior benchmarks and VOICENET-EMO, talking-style attributes for VOICENET-EXT. Real = recorded human speech; In-the-wild = unscripted, naturalistic. Open license means CC-BY or CC-BY-NC 4.0 or equivalent; var. means varies across pooled corpora.

Dataset	Open Licence	Size (#Utts/Hours)	#Labels	#Spk.	Real / In-wild	Multilin.
IEMOCAP (Busso et al., 2008)	✗	10k / ~12h	9	10 (5M/5F)	R / ✗	✗
RAVDESS (Livingstone & Russo, 2018)	✓	1.4k / ~1h	8	24 (12M/12F)	R / ✗	✗
SAVEE (Jackson & Haq, 2014)	✗	480 / <1h	7	4 (Male)	R / ✗	✗
EmoDB (Burkhardt et al., 2005)	✗	535 / <1h	7	10 (5M/5F)	R / ✗	✗
CREMA-D (Cao et al., 2014)	✓	7.4k / ~6h	6	91 (48M/43F)	R / ✗	✗
SERAB (Scheidwasser-Clow et al., 2022)	✗	9 corpora / var.	6	var.	R / ✗	✓
EmoBox (Ma et al., 2024)	✗	32 corpora / var.	≤8	var.	R / partial	✓
SER Evals (Osman et al., 2024)	✗	18 corpora / var.	≤8	var.	R / partial	✓
MSP-Podcast (Lotfian & Busso, 2019)	✓	~100k / ~100h	4	var.	R / ✓	✗
BERSt (Tuttsi et al., 2026)	✓	~4h	6	98	R / ✗	✗
EMONET-VOICE Bench (Schuhmann et al., 2025a)	✓	~12k / ~36h	40	11 (Synth)	Synth / ✗	✓
ours VOICENET-EMO (bench)	✓	7,988 pairs / 3,944 clips	40	var.	R / ✓	✓
VOICENET-EXT (bench)	✓	18.5k pairs	57	var.	R / ✓	✓

A.2 EMOLIA and EMOLIA-BALANCED Corpora

EMOLIA is the fully emotion-annotated version of the Emilia-Large corpus (71.78M clips, 215,600 hours; Emilia portion CC-BY-NC-4.0, Emilia-YODAS portion CC-BY-4.0), annotated with 40 emotion scores, emotion captions, and speaker-timbre embeddings. EMOLIA-BALANCED is a 5.26M-clip balanced subset derived from EMOLIA. Table 4 lists the four corpora that received dense MOSS-Audio annotations.

Table 4: The four MOSS-AUDIO-annotated voice corpora. Each clip is queried with 18 prompt-groups; the per-corpus annotation-call counts follow.

Source dataset	Clips	Annotation calls (18 ×)
Multilingual In The Wild	721k	12.98M
Majestrino	973k	17.51M
LAION’s Got Talent	1.69M	30.34M
EMOLIA-BALANCED (balanced subset of EMOLIA)	5.26M	94.62M
Total	8.64M	155.46M

A.3 EMONET-VOICE Taxonomy

The 40 emotion categories used in EMONET-VOICE, adapted from EMONET-FACE (Schuhmann et al., 2025b), are listed below with associated descriptive terms used during conceptualization and prompting:

- **Amusement:** 'lighthearted fun', 'amusement', 'mirth', 'joviality', 'laughter', 'playfulness', 'silliness', and 'jesting'
- **Elation:** 'happiness', 'excitement', 'joy', 'exhilaration', 'delight', 'jubilation', 'bliss', and 'Cheerfulness'
- **Pleasure/Ecstasy:** 'ecstasy', 'pleasure', 'bliss', 'rapture', and 'Beatitude'
- **Contentment:** 'contentment', 'relaxation', 'peacefulness', 'calmness', 'satisfaction', 'Ease', 'Serenity', 'fulfillment', 'gladness', 'lightness', 'serenity', and 'tranquility'
- **Thankfulness/Gratitude:** 'thankfulness', 'gratitude', 'appreciation', and 'gratefulness'
- **Affection:** 'sympathy', 'compassion', 'warmth', 'trust', 'caring', 'Clemency', 'forgiveness', 'Devotion', 'Tenderness', and 'Reverence'
- **Infatuation:** 'infatuation', 'having a crush', 'romantic desire', 'fondness', 'butterflies in the stomach', and 'adoration'
- **Hope/Enthusiasm/Optimism:** 'hope', 'enthusiasm', 'optimism', 'Anticipation', 'Courage', 'Encouragement', 'Zeal', 'fervor', 'inspiration', and 'Determination'
- **Triumph:** 'triumph', 'superiority'
- **Pride:** 'pride', 'dignity', 'self-confidently', 'honor', and 'self-consciousness'
- **Interest:** 'interest', 'fascination', 'curiosity', and 'intrigue'
- **Awe:** 'awe', 'awestruck', and 'wonder'
- **Astonishment/Surprise:** 'astonishment', 'surprise', 'amazement', 'shock', and 'startlement'
- **Concentration:** 'concentration', 'deep focus', 'engrossment', 'absorption', and 'attention'
- **Contemplation:** 'contemplation', 'thoughtfulness', 'pondering', 'reflection', 'meditation', 'Brooding', and 'Pensiveness'
- **Relief:** 'relief', 'respite', 'alleviation', 'solace', 'comfort', and 'liberation'
- **Longing:** 'yearning', 'longing', 'pining', 'wistfulness', 'nostalgia', 'Craving', 'desire', 'Envy', 'homesickness', and 'saudade'
- **Teasing:** 'teasing', 'bantering', 'mocking playfully', 'ribbing', and 'provoking lightly'
- **Impatience and Irritability:** 'impatience', 'irritability', 'irritation', 'restlessness', 'short-temperedness', and 'exasperation'
- **Sexual Lust:** 'sexual lust', 'carnal desire', 'lust', 'feeling horny', and 'feeling turned on'
- **Doubt:** 'doubt', 'distrust', 'suspicion', 'skepticism', 'uncertainty', and 'Pessimism'
- **Fear:** 'fear', 'terror', 'dread', 'apprehension', 'alarm', 'horror', 'panic', and 'nervousness'
- **Distress:** 'worry', 'anxiety', 'unease', 'anguish', 'trepidation', 'Concern', 'Upset', 'pessimism', and 'foreboding'
- **Confusion:** 'confusion', 'bewilderment', 'flabbergasted', 'disorientation', and 'Perplexity'
- **Embarrassment:** 'embarrassment', 'shyness', 'mortification', 'discomfiture', 'awkwardness', and 'Self-Consciousness'
- **Shame:** 'shame', 'guilt', 'remorse', 'humiliation', and 'contrition'
- **Disappointment:** 'disappointment', 'regret', 'dismay', 'letdown', and 'chagrin'
- **Sadness:** 'sadness', 'sorrow', 'grief', 'melancholy', 'Dejection', 'Despair', 'Self-Pity', 'Sullenness', 'heartache', 'mournfulness', and 'misery'
- **Bitterness:** 'resentment', 'acrimony', 'bitterness', 'cynicism', and 'rancor'
- **Contempt:** 'contempt', 'disapproval', 'scorn', 'disdain', 'loathing', and 'Detestation'
- **Disgust:** 'disgust', 'revulsion', 'repulsion', 'abhorrence', and 'loathing'
- **Anger:** 'anger', 'rage', 'fury', 'hate', 'irascibility', 'enragement', 'Vexation', 'Wrath', 'Peevishness', and 'Annoyance'
- **Malevolence/Malice:** 'spite', 'sadism', 'malevolence', 'malice', 'desire to harm', and 'schadenfreude'
- **Sourness:** 'sourness', 'tartness', 'acidity', 'acerbity', and 'sharpness' (Note: Primarily gustatory, vocal correlates might be subtle reactions)
- **Pain:** 'physical pain', 'suffering', 'torment', 'ache', and 'agony'
- **Helplessness:** 'helplessness', 'powerlessness', 'desperation', and 'submission'
- **Fatigue/Exhaustion:** 'fatigue', 'exhaustion', 'weariness', 'lethargy', 'burnout', and 'Weariness'
- **Emotional Numbness:** 'numbness', 'detachment', 'insensitivity', 'emotional blunting', 'apa-

thy’, ‘existential void’, ‘boredom’, ‘stoicism’, and ‘indifference’

- **Intoxication/Altered States of Consciousness:** ‘being drunk’, ‘stupor’, ‘intoxication’, ‘disorientation’, and ‘altered perception’
- **Jealousy & Envy:** ‘jealousy’, ‘envy’, and ‘covetousness’

A.4 VOICENET-EXT Attribute List

The 57 talking-style attributes annotated in VOICENET-EXT are listed below in groups. Each attribute is shown with its MOSS-Audio short code in `typewriter` (4-letter, with an `R_` prefix for resonance placement and an `S_` prefix for style descriptors) followed by a plain-text description in parentheses; this is the schema used internally by the annotation pipeline (§3.2).

Perceived speaker. `GEND` (gender presentation) and `AGEV` (age range).

Affective dimensions. `VALN` (valence), `AROU` (arousal), `VOLT` (volatility), `VALS` (valence sharpness), `TENS` (vocal tension), `ARSH` (acoustic harshness), and `VULN` (vulnerability).

Prosodic delivery. `TEMP` (speaking rate), `ATCK` (attack), `CHNK` (chunking), `RANG` (melodic range), `VFLX` (vocal flexibility), `STNC` (vocal stance), `EMPH` (emphasis), `DFLU` (disfluency), `CLRT` (clarity), `STRU` (discourse structure), `COGL` (cognitive load), and `FOCS` (focus).

Vocal quality. `ROUG` (roughness), `SMTH` (smoothness), `BRGT` (brightness), `WARM` (warmth), `FULL` (fullness), `HARM` (harmonicity), `METL` (metallicity), `ESTH` (esthetics), `REGS` (register), `RESP` (respiration), `DARC` (dark/light), and `RCQL` (recording quality).

Resonance placement. `R_CHST` (chest), `R_THRT` (throat), `R_ORAL` (oral), `R_HEAD` (head), `R_MASK` (mask), `R_MIXD` (mixed), and `R_NASL` (nasal).

Recording context. `BKGN` (background noise), `EXPL` (expletive content).

Style descriptors. `S_CONV` (conversational), `S_CASU` (casual), `S_PLAY` (playful), `S_CART` (cartoonish), `S_FORM` (formal), `S_AUTH` (authoritative), `S_TECH` (teacher/didactic), `S_MONO` (monologue), `S_DRAM` (dramatic), `S_NARR` (narrator), `S_STRY` (storytelling), `S_NEWS` (newsreader), `S_RANT` (ranting), `S_WHIS` (whisper-talk), and `S_ASMR` (ASMR).

The MOSS-Audio schema additionally produces `BURST` (vocal-burst presence), `EMO` (fine-grained emotion category), `ACNT` (accent), and `LANG` (language), all dropped from VOICENET-EXT because either redundant with VOICENET-EMO, demographic rather than perceptual, or categorical rather than ordinal. The full rubric-level definitions used for human annotation are available at <https://projects.laion.ai/emolia-bench/taxonomy/>.

A.5 Dataset configuration

Both VOICECLAP-SMALL and VOICECLAP-LARGE are trained on the same 9-corpus mixture. Each line of Tab. 5 is one WebDataset entry with clip count and caption-key. EMOLIA-BALANCED, LAION’s Got Talent, and Majestrino supply vocal-style captions; the four FCaps corpora (EARS, Espresso, Voxceleb1, and Voxceleb2) retain their prior FCaps-style captions (Yang et al., 2026b); the two Synthetic Vocal Bursts collections are sourced in-house and carry the same simpler caption schema. Multilingual In The Wild is annotated and released as part of the EMOLIA suite but excluded from the training mixture; an internal ablation showed it did not improve downstream VOICENET-EMO or MAEB-voice scores.

Table 5: VOICECLAP DATA training corpus manifest (Multilingual In The Wild is annotated and released as part of the EMOLIA suite but excluded from training after ablation). Caption-key indicates which JSON field is used as the contrastive text.

Corpus	Clips	Caption key
LAION’s Got Talent	1,685,809	detailed_caption
EMOLIA-BALANCED	5,256,683	emotion_caption
Majestrino	972,658	caption
Synthetic Vocal Bursts	325,548	text (in-house, YouTube)
Improved Synthetic Vocal Bursts	15,680	text (in-house, YouTube)
EARS	17,227	text (FCaps)
Espresso	27,056	text (FCaps)
Voxceleb1	153,516	text (FCaps)
Voxceleb2	600,064	text (FCaps)

A.6 Evaluation Harness and Loader Audit

The harness implementation lives in `open_clap_scaling/eval_emolia_current_{clap,ensemble,baselines}.py`. The function `evaluate_subset` (`eval_emolia_current_ensemble.py`) computes the similarity matrix, sweeps thresholds, and emits per-tag files alongside per-prompt and per-task-type CSVs. Per-prompt thresholds (`find_per_prompt_thresholds`) require at least 10 rows per prompt; otherwise the global threshold is used.

A.7 Per-tag Summary Files

Full per-tag files (with *bal@0*, *bal@opt*, *bal@per_prompt*, mean per-prompt ρ , similarity-distribution stats, and per-prompt thresholds) are released alongside the paper. Numbers reported in this paper match those files exactly. Per-tag files additionally contain the per-(audio, prompt) cosine similarities and binary labels used for Tab. 6. We draw 2,000 paired bootstrap samples with audio clip as the cluster, using the same sampled clips for every model. The 40 emotion and 395 retained attribute-level prompts are fixed; ρ is recomputed within each prompt and then averaged. For *bal@pp*, the Tab. 1 thresholds are locked, so these intervals isolate clip-sampling uncertainty rather than refitting an oracle threshold in every replicate. The released script records the seed and input checksums and also emits paired intervals for every point-sorted adjacent model difference.

Table 6: Paired clip-bootstrap 95% CIs for Tab. 1 (2,000 replicates). Entries are point estimate [lo, hi]. Per-prompt thresholds are held fixed; intervals therefore do not correct their same-set fitting. VOICENET-EXT remains preliminary because this resampling also does not propagate annotation uncertainty.

Model	VOICENET-EMO <i>bal@pp</i>	VOICENET-EXT <i>bal@pp</i>	VOICENET-EMO ρ	VOICENET-EXT ρ
LAION-CLAP	0.5624 [0.5511, 0.5727]	0.6184 [0.6104, 0.6268]	0.0310 [0.0071, 0.0537]	0.0600 [0.0421, 0.0753]
MGA-CLAP	0.5897 [0.5789, 0.5998]	0.6208 [0.6128, 0.6287]	0.0947 [0.0717, 0.1186]	0.0819 [0.0653, 0.0962]
GLAP	0.5765 [0.5661, 0.5877]	0.6254 [0.6171, 0.6341]	0.0667 [0.0425, 0.0907]	0.0836 [0.0655, 0.0992]
M2D-CLAP-2025	0.5681 [0.5571, 0.5787]	0.6361 [0.6273, 0.6444]	0.0348 [0.0121, 0.0582]	0.1003 [0.0822, 0.1142]
CLSP	0.5675 [0.5563, 0.5791]	0.6221 [0.6140, 0.6303]	0.0610 [0.0380, 0.0840]	0.0937 [0.0763, 0.1072]
MS-CLAP-2023	0.5754 [0.5646, 0.5858]	0.6321 [0.6243, 0.6403]	−0.0084 [−0.0303, 0.0153]	0.1044 [0.0867, 0.1186]
Cacophony	0.5432 [0.5329, 0.5535]	0.6089 [0.6005, 0.6173]	−0.0183 [−0.0414, 0.0034]	0.0188 [0.0021, 0.0350]
LCO-Embedding-Omni-7B	0.6632 [0.6525, 0.6740]	0.6342 [0.6259, 0.6419]	0.3052 [0.2824, 0.3266]	0.0977 [0.0800, 0.1121]
LCO-Embedding-Omni-3B	0.6547 [0.6435, 0.6656]	0.6293 [0.6210, 0.6375]	0.2876 [0.2646, 0.3082]	0.0873 [0.0703, 0.1018]
VOICECLAP-SMALL	0.6754 [0.6648, 0.6860]	0.6367 [0.6289, 0.6453]	0.3176 [0.2932, 0.3391]	0.1051 [0.0867, 0.1189]
VOICECLAP-LARGE	0.7021 [0.6914, 0.7118]	0.6510 [0.6431, 0.6588]	0.3719 [0.3487, 0.3923]	0.1475 [0.1292, 0.1599]

A.8 VOICENET-EMO Per-Emotion ρ

Table 7: Per-emotion Spearman ρ on VOICENET-EMO for VOICECLAP, the Omni-Embedding bases, and all seven general-audio CLAP baselines. Rows are sorted by the mean ρ across the displayed model columns; the mean is used only for ordering and is not shown. Best per row is bold, runner-up underlined. Cells are coloured by ρ .

emotion	VOICECLAP Small	VOICECLAP Large	LCO Omni-3B	LCO Omni-7B	LAION CLAP	MS-CLAP 2023	CLSP	GLAP	M2D CLAP	MGA CLAP	Cacophony
Anger	0.555	0.577	0.455	0.524	0.310	0.348	0.486	0.389	0.228	0.448	0.054
Impatience and Irritability	0.638	<u>0.581</u>	0.524	0.544	0.212	-0.086	0.149	0.375	-0.132	0.489	0.092
Contempt	0.398	<u>0.441</u>	0.433	0.497	0.230	-0.008	0.149	0.254	0.186	0.394	0.007
Amusement	0.518	0.500	0.509	0.530	0.283	-0.281	-0.351	<u>0.530</u>	0.062	0.526	0.154
Distress	0.627	<u>0.626</u>	0.538	0.575	-0.030	-0.255	0.226	0.226	0.325	0.382	-0.318
Contemplation	0.449	0.464	0.437	0.423	0.022	0.312	0.161	-0.022	0.138	0.161	0.185
Disgust	0.313	<u>0.513</u>	0.497	0.528	-0.031	0.135	0.172	0.077	0.165	0.278	-0.027
Helplessness	<u>0.444</u>	0.508	0.382	0.418	-0.262	-0.015	0.257	0.250	0.240	0.351	-0.214
Malevolence/Malice	0.512	<u>0.506</u>	0.424	0.482	0.153	-0.015	-0.109	0.179	0.075	0.063	-0.146
Sadness	<u>0.533</u>	0.569	0.381	0.443	-0.083	0.212	0.150	-0.167	0.206	-0.042	-0.194
Confusion	0.365	0.429	0.386	<u>0.391</u>	-0.063	-0.190	-0.017	0.276	-0.004	0.215	0.105
Teasing	0.132	0.370	<u>0.413</u>	0.501	0.265	-0.276	-0.150	0.031	-0.049	0.369	0.228
Bitterness	0.361	0.314	0.158	0.255	0.234	0.166	0.171	-0.002	0.169	0.175	-0.198
Embarrassment	0.320	0.325	0.265	0.240	0.053	-0.049	0.178	0.187	-0.002	0.192	0.042
Triumph	0.292	0.363	0.291	0.271	0.149	0.065	-0.108	0.138	-0.030	0.205	0.087
Relief	0.279	0.415	0.279	<u>0.328</u>	-0.122	0.083	<u>0.251</u>	-0.056	0.133	0.046	0.046
Disappointment	0.321	0.377	0.386	0.368	-0.071	0.180	0.116	0.058	0.166	-0.147	-0.093
Fatigue/Exhaustion	0.489	0.505	0.362	0.347	-0.135	-0.054	-0.028	0.082	0.155	0.009	-0.083
Thankfulness/Gratitude	0.068	0.252	0.351	<u>0.307</u>	-0.003	0.020	0.135	0.136	0.164	0.119	0.010
Pride	<u>0.268</u>	0.279	0.190	0.179	-0.024	0.057	0.171	0.108	0.070	0.145	-0.029
Affection	0.523	<u>0.501</u>	0.462	0.473	-0.047	-0.069	0.109	-0.024	-0.113	-0.259	-0.157
Pleasure/Ecstasy	0.411	<u>0.516</u>	0.298	0.399	0.062	-0.388	0.195	0.069	-0.096	-0.059	-0.037
Pain	0.358	0.364	0.316	0.326	-0.164	0.087	0.115	0.048	0.118	-0.043	-0.194
Awe	0.434	0.472	<u>0.436</u>	0.378	-0.143	0.097	-0.112	-0.159	-0.025	-0.134	-0.054
Intoxication/Altered States	0.178	0.243	<u>0.238</u>	0.227	-0.008	-0.177	0.095	0.144	0.151	0.061	0.017
Longing	0.455	<u>0.400</u>	0.370	0.324	0.022	-0.218	-0.014	-0.028	-0.067	-0.140	0.025
Concentration	0.271	0.094	-0.041	-0.013	0.122	<u>0.235</u>	0.092	0.095	-0.175	0.229	0.061
Elation	0.347	0.408	<u>0.406</u>	<u>0.407</u>	0.136	-0.372	-0.205	0.136	-0.290	-0.224	0.157
Doubt	-0.038	0.132	0.047	0.064	0.136	-0.051	-0.026	0.141	0.287	<u>0.169</u>	0.014
Sourness	-0.028	<u>0.187</u>	0.093	0.149	0.048	0.051	0.113	-0.060	0.047	0.200	0.017
Fear	<u>0.200</u>	0.251	0.133	0.165	-0.095	0.102	0.037	-0.071	-0.060	0.128	-0.044
Contentment	0.295	0.401	0.286	<u>0.305</u>	0.017	0.068	-0.120	-0.239	-0.285	-0.181	0.161
Hope/Enthusiasm/Optimism	0.255	<u>0.263</u>	0.268	0.221	-0.156	-0.157	-0.158	0.030	-0.042	-0.023	0.181
Infatuation	<u>0.150</u>	0.286	0.137	0.114	-0.071	-0.016	0.109	-0.041	-0.018	0.012	-0.022
Interest	0.046	0.057	0.124	0.163	-0.105	0.002	-0.059	0.094	0.074	<u>0.154</u>	0.037
Shame	0.004	0.285	0.249	<u>0.260</u>	0.055	-0.127	-0.028	0.122	-0.147	-0.043	-0.122
Jealousy/Envy	0.137	0.239	0.093	<u>0.168</u>	0.091	-0.079	-0.064	-0.093	0.075	-0.067	-0.095
Emotional Numbness	0.459	0.476	-0.233	-0.198	0.322	<u>0.472</u>	0.234	-0.430	-0.180	-0.278	-0.300
Sexual Lust	0.109	0.187	<u>0.143</u>	0.128	-0.071	0.046	0.035	-0.037	-0.018	-0.076	-0.107
Astonishment/Surprise	0.258	<u>0.200</u>	0.017	-0.001	-0.001	-0.189	0.083	-0.082	-0.105	-0.018	0.023

A.9 VOICENET-EXT Per-Attribute ρ

Per-attribute Spearman ρ for VOICENET-EXT covers 57 attributes $\times$ 7 levels (399 (attribute, level) prompts). The full per-attribute ρ table is released alongside the paper rather than reproduced here, because the averages quoted in the main text ($\rho=0.15$ for VOICECLAP-LARGE, $\rho=0.11$ for VOICECLAP-SMALL) summarise the signal adequately: per-attribute values are noisy at ~ 46 audios per (attribute, level) prompt. We additionally release `per_attribute_reliability.csv`, containing one row per attribute with total and per-rater-count coverage, majority-yes prevalence and Wilson interval, strict three-rater observed agreement and Fleiss κ , item-bootstrap 95% intervals, and reliable-core membership. Its aggregate exact estimate is $\kappa = -0.0218$ with a 95% item-bootstrap interval $[-0.0367, -0.0066]$, which excludes zero. The accompanying script fixes the bootstrap seed and records the source-annotation SHA-256. Tab. 8 records the underlying coverage that each released ρ value is computed from — the per-attribute pair count n and the human-vote distribution — so readers can see at a glance which attributes are well-sampled (e.g. S_RANT, ARSH, S_DRAM at $n=350$) and which are sparser (FULL at 273 pairs, EXPL at 144).

A.10 Human Audit of the MOSS-Audio Labels (Interim)

Because the dense MOSS-AUDIO annotations that supervise VOICECLAP are model-generated, we run a prespecified audit of how closely a human rater reproduces them. The design was fixed in advance: 300 items, 75 English clips drawn from each of the four MOSS-annotated corpora, one attribute per clip, all 59 prompt-group attributes represented, plus 24 hidden checks that pair a clip with a genuine MOSS sentence written for a different clip at a distant level of the same attribute. The rater first selects a rubric level independently, and only then is the candidate MOSS label revealed and judged for fit, so the exact-level statistic cannot be anchored on the machine’s answer. The sample is English-only so that every attribute is answerable and comprehension never confounds the judgement; it therefore measures annotation quality on the English portion of corpora that ship multilingual. Since the corpora store one free-text sentence per (clip, attribute) and no numeric level, each sentence was mapped back onto its rubric by a text-only pass run with the anchors in both directions, and only items on which both directions agree carry a level and enter the level-match statistic.

The audit is not finished. Five raters have contributed 310 judgements covering 97 of the 300 items, and no rater has completed a full pass, so intervals resample items rather than judgements and the result below is interim. Raters chose the same rubric level as MOSS on 0.257 of mapped judgements (95% CI [0.191, 0.325]) against a permutation baseline of 0.169, and within one level on 0.563 (95% CI [0.485, 0.646]) against 0.473. Because several raters judged the same items, the identical statistic is computable between raters: two raters chose the same level on 0.275 (95% CI [0.220, 0.337]). The paired difference between human–human and human–MOSS agreement is +0.018 (95% CI [−0.053, +0.102]) for exact match and +0.087 (95% CI [−0.010, +0.192]) within one level; both contain zero. The rank correlation between the MOSS level and the rater level is 0.237, which falls between the 50th and 75th percentile of the effect-size scale discussed in §5 (Gignac & Szodorai, 2016).

Two properties of the instrument belong alongside those numbers. Raters were also asked whether the MOSS sentence fits the clip and accepted it in 0.732 of judgements, but only 2 of 12 hidden checks were rejected, while on those checks their own level choice sat 2.2 levels from the planted label against 1.5 on real items. Raters located the discrepancy and then declined to call it a mismatch, so the fit judgement is too permissive to serve as the primary measure and exact level match is used instead. The sample supports overall statements only, not per-attribute ones. The audit set, the per-item judgements, and the analysis script are available at <https://github.com/LAION-AI/emolia-bench>; the completed audit will be reported whatever it shows.

A.11 Detailed Taxonomy Construction Methodology

The 40-category emotion taxonomy utilized in both the EMONET-VOICE foundation and benchmark datasets was originally developed for the EmoNet-Face Benchmark (Schuhmann et al., 2025b).

The taxonomy aims to support fine-grained understanding of affective states beyond basic-emotion models, drawing on the Theory of Constructed Emotion (TCE) (Barrett, 2017).

The taxonomy spans positive and negative emotions, social emotions (*Embarrassment*, *Shame*, and *Pride*), cognitive states (*Concentration*, *Doubt*, and *Confusion*), bodily states (*Pain*, *Fatigue*, and *Intoxication*), and less common categories such as *Sourness* and *Helplessness*. The full list of 40 categories with descriptive word clusters appears in App. A.3; the same list is used in the prior EmoNet-Face paper.

The construction process involved several key stages:

1. **Literature-Driven Candidate Extraction:** The comprehensive "Handbook of Emotions" (946 pages) (Lewis et al., 2010) was digitized using Optical Character Recognition (OCR). The digitized text was then divided into manageable 500-word segments.

2. **AI-Assisted Term Identification:** GPT-4 was employed to analyze these text segments and extract potential nouns representing emotion concepts.
3. **Refinement and Deduplication:** The initially extracted terms were aggregated, and duplicates were removed, resulting in a candidate list of approximately 170 unique emotion-related nouns.
4. **Expert-Guided Clustering and Categorization:** This refined list of terms underwent an iterative process of clustering. This involved independent categorization efforts by the taxonomy authors, followed by critical reviews and discussions. Psychologists and researchers in affective computing provided expert guidance throughout this phase to ensure the semantic coherence and psychological relevance of the emerging categories. Each of the final 40 categories represents a cluster of these semantically related emotion words.

In line with the Theory of Constructed Emotion, this taxonomy does not presuppose the biological universality or fixedness of these emotional categories. Instead, it is intended to facilitate context-aware and socially informed interpretations of affective expressions by AI systems. Recognizing the inherent ambiguity in perceiving emotions (e.g., a high-arousal vocal expression might be interpreted as amusement, elation, or excitement depending on context and observer), the taxonomy was specifically designed to support plausible multi-label annotations rather than forcing rigid, single-label classifications. This approach aims to enable richer and more contextually sensitive representations of emotion in AI.

A.12 Per-Emotion Annotator Agreement on VOICENET-EMO

Fig. 1 reports the distribution of binary (present vs. absent) three-rater outcomes per emotion on the 7,984 fully rated VOICENET-EMO pairs. For each emotion, the stacked bar shows the fraction of pairs on which all three experts mark the emotion present (3:0 unanimous present), two of three mark it present (2:1), one of three marks it present (1:2), or none does (3:0 unanimous absent). Across all pairs, the three experts are unanimous on 32.7% of items. Unanimity is highest for *Thankfulness/Gratitude* (68%), *Interest* (60%), *Hope/Enthusiasm/Optimism* (57%), and *Doubt* (55%), almost entirely through unanimous-present votes, and lowest for *Intoxication/Altered States* (19%), *Infatuation* (21%), and *Teasing, Pride, and Longing* (24% each). Per-emotion unanimity is unrelated to per-emotion model performance in Tab. 7: across the 40 emotions its Spearman correlation with VOICECLAP-LARGE ρ is 0.02, and *Interest* and *Doubt* combine high unanimity with some of the weakest model signal. This matches the agreement-stratified analysis in §4.3, which finds no concentration of model gains in higher-agreement emotions.

A.13 VOICECLAP Training Details

VOICECLAP-SMALL. A 110M-parameter dual-tower CLAP (Wu et al., 2023): BUD-E-Whisper-Small (Radford et al., 2023) (768-d audio output) and all-MiniLM-L6-v2 (Reimers & Gurevych, 2019; Wang et al., 2020) with mean-pooling (384-d text output), each with a learnable linear projection to a shared 768-d space. Training uses SigLIP sigmoid contrastive loss (Zhai et al., 2023), AdamW ($\beta_1=0.9$, $\beta_2=0.95$, weight decay 0.2), LR 10^{-4} with 200-step warmup and cosine decay, gradient clip 1.0, per-GPU batch 128 on 4 GH200s (effective batch 512 with all-gather), one epoch (~ 0.5 h).

VOICECLAP-LARGE. A single-tower rank-16 LoRA (Hu et al., 2022) (alpha 32, dropout 0.05) finetune of LCO-EMBEDDING-OMNI-7B (Xiao et al., 2025) (Qwen2.5-Omni-Thinker-7B backbone with sentence-transformer head, 3,584-d output). Training uses symmetric InfoNCE (van den Oord et al., 2018) on cosine similarities at fixed logit scale $\log(1/0.07)$ with all-gather negatives, AdamW ($\beta_2=0.95$, weight decay 0.01), LR 10^{-4} (200-step warmup, cosine decay), gradient clip 1.0. On four GH200s, micro-batch 2 with 16-step accumulation gives a 128 optimizer-step batch and an effective contrastive batch of 8; accumulation reduces noise but does not enlarge the InfoNCE negative pool. One epoch (~ 2 h).

A.14 VOICENET-EXT Annotation Coverage and Reliability

VOICENET-EXT asks: *does clip C exhibit attribute A at level L?* for the 57 talking-style attributes of §3.1. The audio source is distinct from EMOLIA-BALANCED. Each of the 57 dimensions is defined by a rubric with ordinal levels (seven levels, 0–6, for nearly all attributes); the full rubric definitions are available at <https://projects.laion.ai/emolia-bench/taxonomy/>. For each (dimension, level) bucket, the annotation task is reduced to a binary confirmation: candidate audio clips are pre-screened by Gemini 3 Flash, which confirms whether each clip matches the target level description. An equal number of positives (clips confirmed to match the target level) and negatives (clips confirmed to match a non-adjacent level of the same dimension) are presented to human annotators, who listen and confirm or reject the match. This design ensures that annotators perform a focused binary judgement rather than a full ordinal rating. The release contains 18.5k (audio, attribute-level) pairs carrying 40,990 ratings from eight annotators; four hold psychology degrees (BSc to MSc, including clinical psychology) and contributed 95.3% of all ratings, while the remaining four come from law, translation, computer science, and environmental-science backgrounds. 5,828 pairs carry at least three ratings (5,583 with exactly three, 241 with four, and 4 with five), 10,338 carry two, and 2,366 carry one. Aggregate majority-yes rate is 0.284. On the 5,583 non-LANG items with exactly three ratings, Fleiss’ binary $\kappa = -0.022$ (95% item-bootstrap CI $[-0.037, -0.007]$) and observed rater-pair agreement is 0.494 $[0.486, 0.501]$; pairwise and leave-one-out balanced-accuracy references are 0.533 and 0.479 (the pairwise figure averages over rater pairs; weighting instead by the 28,573 paired votes gives 0.484). With 5,583 three-rater items the κ interval excludes zero, so near-chance agreement on VOICENET-EXT is a stable property of these fine-grained rubric-level judgements as currently operationalised, not an artefact of sparse annotation. We prespecified a reliable-core rule before computing the per-attribute results: at least 20 three-rater items, $\kappa \geq 0.20$, and a 95% lower bound above zero. Every attribute clears the sample-size bar (all 57 have $n \geq 20$ strict three-rater items, median $n=98$), but none reaches $\kappa \geq 0.20$: the maximum is STNC at $\kappa=0.085$ $[-0.043, 0.211]$, only 17 of 57 attributes have $\kappa > 0$, and the median is -0.031 . No attribute therefore enters the uncertainty-supported core. We therefore label all current VOICENET-EXT scores preliminary and treat its attributes as exploratory perceptual probes. Per-attribute prevalence, coverage, agreement, κ , uncertainty, and model ρ files are released alongside the benchmark (App. A.9).

A.15 Additional Results

Human baselines and ceilings. Four reference points anchor the model numbers. Random and always-predict-majority both give 0.500 balanced accuracy by construction (with always-predict-majority reaching 0.578 raw accuracy on VOICENET-EMO and 0.716 on VOICENET-EXT due to the present/absent class skew). The pairwise human ceiling scores rater A’s binary vote against rater B’s on items with ≥ 2 raters: 0.562 balanced accuracy on VOICENET-EMO (24.0k paired votes) and 0.533 on VOICENET-EXT (averaged over rater pairs; 0.484 if weighted by the 28.6k paired votes instead). The leave-one-out ceiling scores one rater against the majority of the others on items with at least three ratings: 0.572 on VOICENET-EMO (13.2k comparisons) and 0.479 on VOICENET-EXT (9,246 LOO comparisons across the 5,828 items with three or more ratings). On VOICENET-EMO, VOICECLAP-LARGE’s 0.7021 *bal@pp* is above the 0.572 LOO reference, i.e. closer to the multi-rater consensus than any individual expert is. This is not a like-for-like comparison—the model is scored against the aggregate majority label while each expert is scored against the other raters—so it should be read as strong alignment with the expert consensus, not as surpassing human emotion perception. On VOICENET-EXT the human ceilings sit at chance level, below our model scores, reflecting both the harder rubric task and the more heterogeneous rater pool. Human consistency on VOICENET-EXT is therefore at chance on the current rubric, which is why we treat its model scores as exploratory separability rather than a validated ceiling comparison.

Calibration. Per-prompt thresholding most benefits models whose similarity distribution is biased away from zero. VOICECLAP-LARGE has *bal@0* = 0.580, *bal@opt* = 0.673 at $\theta=0.136$, and

$bal@pp = 0.702$. The 12-pp gap from $bal@0$ to $bal@pp$ reflects a positive similarity bias (mean cosine ≈ 0.14) and would be largely closed by post-hoc mean-centring. VOICECLAP-SMALL has $bal@0 = 0.659$, $bal@opt = 0.661$ at $\theta = -0.020$, and $bal@pp = 0.675$. Its similarity distribution is well-centred and offers almost no calibration headroom: the $bal@pp$ lead over LCO-Omni-7B (0.6754 vs. 0.6632) shrinks to a near-tie under $bal@0$. The headline ordering between the four families (general-audio CLAPs < Omni bases $\approx$ VOICECLAP-SMALL < VOICECLAP-LARGE) is preserved under $bal@opt$ and per-prompt ρ , but readers comparing two models within the same family should consult App. A.7 for both metrics.

Agreement-stratified results. We split the 40 emotions and 57 non-LANG attributes into equal-size lower- and higher-agreement ranks using observed annotator-pair agreement on strict three-rater items, before inspecting model scores. On VOICENET-EMO, the VOICECLAP-LARGE minus LCO-EMBEDDING-OMNI-7B advantage is $+0.0436$ versus $+0.0306$ $bal@pp$ and $+0.0787$ versus $+0.0548$ ρ in the lower- versus higher-agreement halves. The corresponding high-minus-low interactions are -0.0130 (95% CI $[-0.0350, 0.0076]$) and -0.0239 $[-0.0628, 0.0159]$. On preliminary VOICENET-EXT, the advantages are $+0.0210$ versus $+0.0130$ $bal@pp$ and $+0.0453$ versus $+0.0541$ ρ ; interactions are -0.0080 $[-0.0270, 0.0121]$ and $+0.0088$ $[-0.0274, 0.0481]$. Thus none of the four paired interactions excludes zero: these data do not show that VoiceCLAP’s edge is concentrated in the relatively higher-agreement categories. Here “higher” is only a within-VOICENET-EXT rank, not evidence that those attributes are reliably annotated.

Cross-dataset gap diagnosis. A paired follow-up analysis clarifies the two columns where VOICECLAP-LARGE trails its 7B base. On synthetic EmoNet-Voice, the same-runtime Large-minus-base gap is -0.0115 (95% clip-bootstrap CI $[-0.0172, -0.0058]$), with gains and losses distributed unevenly across emotion classes rather than a uniform regression. On RAVDESS, the micro-accuracy gap is -0.0139 , but its actor-clustered interval includes zero ($[-0.0368, 0.0104]$); Large simultaneously raises macro F1 from 0.285 to 0.330 and macro recall from 0.300 to 0.348, with the micro loss dominated by a calm/neutral boundary shift. Because IEMOCAP, RAVDESS, and CREMA-D all contain acted speech while Large improves on two of them, these results do not support a generic acted-versus-naturalistic explanation. They instead indicate dataset- and taxonomy-specific boundary changes that micro accuracy alone can hide.

Scaling view. Fig. 2 plots the average $bal@pp$ of Tab. 1 against model size.

A.16 Effect Sizes in Psychological Context

Interpreting the correlations in psychological context. To contextualise the ρ values in Tab. 1, we draw on modern benchmarks for effect sizes in psychological research. A large-scale meta-analysis finds that the 25th, 50th, and 75th percentiles for correlations in individual-differences research are $r = .11$, $.19$, and $.29$, respectively (Gignac & Szodorai, 2016). By these standards, the seven general-audio CLAPs (all $|\rho| < 0.1$ on VOICENET-EMO) fall below even the 25th percentile—they carry essentially no emotion-ranking signal. Notably, this includes CLSP (Yang et al., 2026b), a speech-specialised contrastive model built on the SPEAR-XLarge speech encoder (Yang et al., 2025), which achieves only $\rho = 0.061$ on VOICENET-EMO despite being explicitly designed for speech rather than general audio; speech-aware pretraining alone is insufficient without dense vocal-style supervision. The two Omni-Embedding bases ($\rho = 0.29$ – 0.31) approach the 75th percentile of published psychological effect sizes. VOICECLAP-LARGE ($\rho = 0.372$) surpasses this further, and its per-emotion peaks—distress (0.63), impatience (0.58), anger (0.58), sadness (0.57)—reach correlations that are exceedingly rare in psychological research.

These numbers must be read against the fundamental limits of *empathic accuracy*—the ability to correctly judge the thoughts and feelings of others—which is “inherently an interpersonal process that requires active engagement of the emotional systems of both interaction partners” (Lin et al., 2024). Because there is no perfect objective ground truth for perceived emotion intensity, human raters naturally disagree on both category and degree (Rum & Perry, 2020; Zaki et al., 2009). A model

achieving $\rho=0.37$ against aggregated expert ratings on a 40-class taxonomy of naturalistic in-the-wild speech has therefore extracted a robust rank-ordered signal from inherently subjective human data. Modern methodological work further cautions against “heuristically dismissing ‘small’ effects as unimportant” (Anvari et al., 2023), since even moderate correlations carry substantial predictive value in complex real-world dynamics. By these standards, VOICECLAP-LARGE’s emotion-ranking performance is not merely competitive—it represents a strong effect size that substantially exceeds what is typical in published psychological research.

A.17 Extended Limitations

The evaluation is representation-level throughout. All eleven systems are voice-text embedding models scored by cosine similarity, so the results characterise how well such models rank vocal-attribute descriptions; they do not establish that better VOICENET scores translate into better end-to-end voice assistants, conversational turn-taking, or response generation, and no spoken-dialogue system is evaluated here. Generative audio-language models are likewise out of scope, as prompting them for graded attribute presence requires a scoring interface that is not comparable to the contrastive protocol used here. EMOLIA and the three sibling voice corpora carry the source-bias of the open Emilia, LAION’s Got Talent, Majestrino, and Multilingual In The Wild collections. Geographic, demographic, and recording-environment coverage are non-uniform. The released 18-prompt-group MOSS-Audio annotations are themselves model-generated; they reflect one model’s interpretation of vocal style and are not validated against human ratings at scale. A prespecified human audit of these labels is underway and reported as an interim result in App. A.10: raters select a rubric level independently before the MOSS label is revealed, and they match it exactly on 0.257 of mapped judgements (95% CI [0.191, 0.325]) against a permutation baseline of 0.169. On the same items two raters match each other on 0.275 (95% CI [0.220, 0.337]), so the paired human–MOSS versus human–human difference is +0.018 (95% CI [−0.053, +0.102]) and contains zero. A rater therefore agrees with the MOSS level about as closely as two raters agree with one another, which bounds the annotation bias rather than establishing that the labels are correct; the audit covers 97 of 300 items with no complete rater pass, and the completed result will be reported whatever it shows. VOICENET-EXT is rated by a heterogeneous eight-person pool (four with psychology degrees, contributing 95.3% of ratings). 5,828 of its 18.5k items carry at least three ratings, but its aggregate κ is -0.022 with a 95% interval [−0.037, −0.007] that now excludes zero from below, and none of the 57 attributes passes our reliable-core rule. Its present model scores are therefore preliminary and should not be used for confirmatory model ranking. Because agreement is at chance even at this coverage, additional ratings alone are unlikely to stabilise the VOICENET-EXT rankings; a rubric revision with anchor-example rater training is required, and until then the subset is best read as an exploratory probe. Within the 40-emotion taxonomy itself, several categories elicit only weak model signal across all eleven evaluated systems — *Sourness* (taxonomy notes its primarily gustatory origin), *Jealousy/Envy*, *Doubt*, *Interest*, and *Sexual Lust* stay at $|\rho| \leq 0.29$ for the best column. We retain them in the released benchmark because removing them would silently shrink the taxonomy, but downstream users may wish to mask them when computing aggregate scores. The bootstrap CIs in Tab. 6 quantify clip-sampling uncertainty conditional on the aggregate labels and the per-prompt thresholds fitted for Tab. 1; they neither correct the same-set threshold optimism nor propagate rater uncertainty. Although VOICECLAP-LARGE’s paired lead is robust on all four columns, most smaller adjacent-model differences overlap zero. Four same-runtime VOICECLAP-SMALL runs (the original seed and three exact-recipe replications) give VOICENET-EMO $bal@pp$ 0.6701 ± 0.0081 and ρ 0.3122 ± 0.0058 , and preliminary VOICENET-EXT $bal@pp$ 0.6327 ± 0.0079 and ρ 0.0986 ± 0.0303 (mean $\pm$ sample standard deviation); this is a descriptive variance proxy for Small, not evidence of seed stability for VOICECLAP-LARGE, whose 7B-scale retraining remains infeasible within our compute budget. VOICECLAP-SMALL and VOICECLAP-LARGE are trained on a mixture of nine voice corpora: EMOLIA-BALANCED, LAION’s Got Talent, Majestrino, two in-house Synthetic Vocal Bursts collections, and four FCaps-captioned corpora (EARS, Espresso, Voxceleb1, and Voxceleb2). Multilingual In The Wild is annotated and released as part of the EMOLIA suite, but did not improve

downstream scores in an internal ablation and is therefore excluded from the released training mixture. Finally, two in-house Synthetic Vocal Bursts collections (341k clips) were curated from Creative Commons-licensed YouTube uploads; we will honour all takedown requests against the released bundle.

B Annotation Platform Instructions and UI

Annotation platform screenshots for both benchmark subsets are shown below. The first annotation interface (Figs. 3–4) was used for the expert annotation of VOICENET-EMO (40-emotion rating). The second annotation interface (Figs. 5–6) was used for the binary level confirmation task on VOICENET-EMO and VOICENET-EXT, respectively.

Table 8: Per-attribute coverage and majority-vote statistics on VOICENET-EXT. n is the number of (audio, attribute-level) pairs released for that attribute, aggregated across the seven rubric levels; per-(attribute, level) prompt counts are therefore roughly $n/7$. *maj-yes* is the fraction of items where the rater majority voted “present”; *polar. match* is the fraction of items where the majority vote agrees with the prompt’s intended affirmative-vs.-contrastive polarity. Per-attribute Spearman ρ for each model is computed over the same n items per attribute and is released alongside the paper as a data artifact. Attribute codes follow the schema in App. A.4; rows are ordered to match that grouping.

Code	Description	n	maj-yes	polar. match
<i>Perceived speaker (2)</i>				
GEND	gender presentation	350	0.237	0.594
AGEV	age range	349	0.284	0.564
<i>Affective dimensions (7)</i>				
VALN	valence	343	0.315	0.624
AROU	arousal	343	0.280	0.618
VOLT	volatility	318	0.343	0.566
VALS	valence sharpness	328	0.216	0.573
TENS	vocal tension	302	0.288	0.583
ARSH	acoustic harshness	350	0.206	0.574
VULN	vulnerability	350	0.294	0.571
<i>Prosodic delivery (12)</i>				
TEMP	speaking rate	318	0.236	0.531
ATCK	attack	350	0.357	0.623
CHNK	chunking	321	0.280	0.523
RANG	melodic range	302	0.288	0.540
VFLX	vocal flexibility	350	0.174	0.531
STNC	vocal stance	342	0.310	0.640
EMPH	emphasis	313	0.339	0.607
DFLU	disfluency	340	0.332	0.647
CLRT	clarity	332	0.310	0.572
STRU	discourse structure	305	0.334	0.626
COGL	cognitive load	320	0.328	0.600
FOCS	focus	341	0.238	0.560
<i>Vocal quality (12)</i>				
ROUG	roughness	303	0.267	0.587
SMTH	smoothness	314	0.287	0.564
BRGT	brightness	320	0.291	0.569
WARM	warmth	314	0.255	0.599
FULL	fullness	273	0.245	0.560
HARM	harmonicity	313	0.198	0.556
METL	metallicity	322	0.261	0.581
ESTH	esthetics	317	0.322	0.590
REGS	register	350	0.211	0.574
RESP	respiration	344	0.334	0.616
DARC	dark/light	329	0.216	0.568
RCQL	recording quality	286	0.329	0.451
<i>Resonance placement (7)</i>				
R_CHST	chest	320	0.316	0.537
R_THRT	throat	336	0.271	0.583
R_ORAL	oral	311	0.280	0.566
R_HEAD	head	340	0.271	0.503
R_MASK	mask	332	0.304	0.596
R_MIXD	mixed	318	0.252	0.560
R_NASL	nasal	279	0.226	0.595
<i>Recording context (2)</i>				
BKGN	background noise	315	0.273	0.546
EXPL	expletive content	144	0.361	0.660
<i>Style descriptors (15)</i>				
S_CONV	conversational	333	0.288	0.643
S_CASU	casual	316	0.282	0.620
S_PLAY	playful	350	0.254	0.594
S_CART	cartoonish	347	0.303	0.631
S_FORM	formal	342	0.319	0.643
S_AUTH	authoritative	338	0.302	0.660
S_TECH	teacher/didactic	322	0.323	0.575
S_MONO	monologue	350	0.371	0.609
S_DRAM	dramatic	350	0.260	0.640
S_NARR	narrator	329	0.307	0.614
S_STRY	storytelling	335	0.346	0.648
S_NEWS	newsreader	340	0.315	0.647
S_RANT	ranting	350	0.266	0.640
S_WHIS	whisper-talk	350	0.300	0.651
S_ASMR	ASMR	333	0.255	0.625
Total / mean	57 attributes	18,532	0.284	0.591

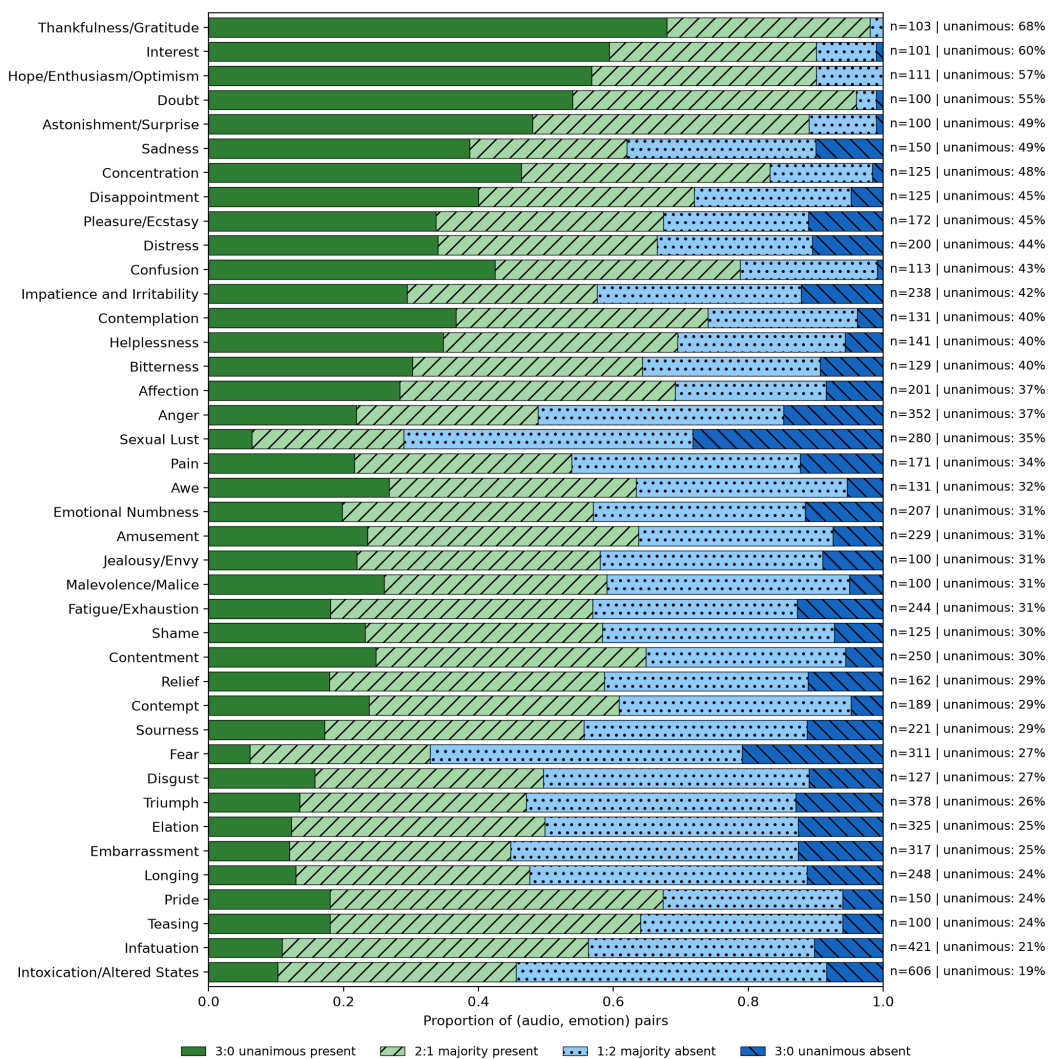


Figure 1: Expert agreement on emotion presence for VOICENET-EMO, per emotion, on the 7,984 pairs with three ratings. Stacked bars show the proportion of (audio, emotion) pairs by the number of experts who marked the emotion present (weakly or strongly): three (3:0 unanimous present), two (2:1), one (1:2), or none (3:0 unanimous absent). The text to the right of each bar gives the number of pairs n and the share of unanimous pairs. Rows are sorted by unanimity.

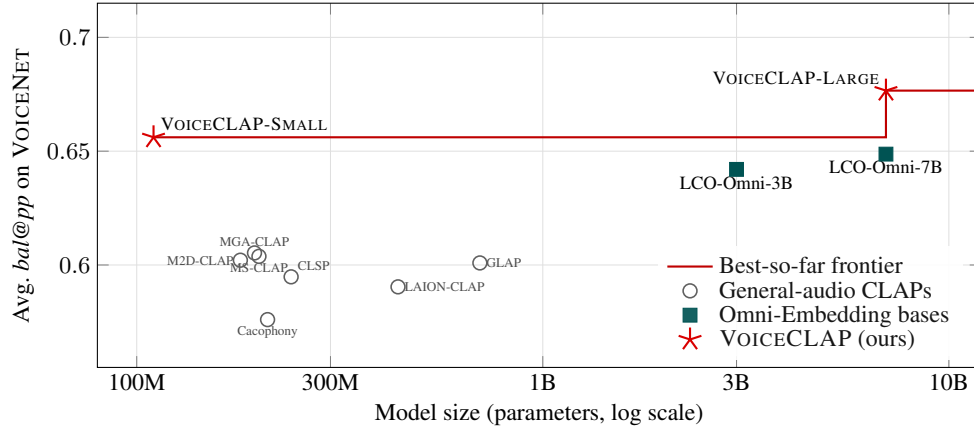


Figure 2: **Scaling view of VOICENET performance.** Each point is one model. The y-axis is the unweighted mean of VOICENET-EMO and VOICENET-EXT per-prompt balanced accuracy from Table 1. The red step line tracks the best score reached by any model up to each parameter count. VOICECLAP-SMALL (110M) sits above the general-audio CLAP cluster and is competitive with the larger Omni bases, while the 7B VOICECLAP-LARGE reaches the highest average score. Parameter counts are approximate (audio + text encoder, before LoRA adapters). Averages match Tab. 1.

Audio Emotion Annotation

Instructions

In this task, you'll be assessing whether a specific emotion appears to be present in the audio recordings. Each recording will be associated with a single emotion label, and you need to decide whether that emotion is:

- **Not Present** - The emotion is not detectable in the audio
- **Weakly Present** - The emotion is somewhat present but not strong
- **Strongly Present** - The emotion is clearly and strongly expressed

Listen carefully to each recording and make your selection based on your perception of the emotion in the audio.

Figure 3: Instructions given to the human annotator for the expert annotation of VOICENET-EMO.

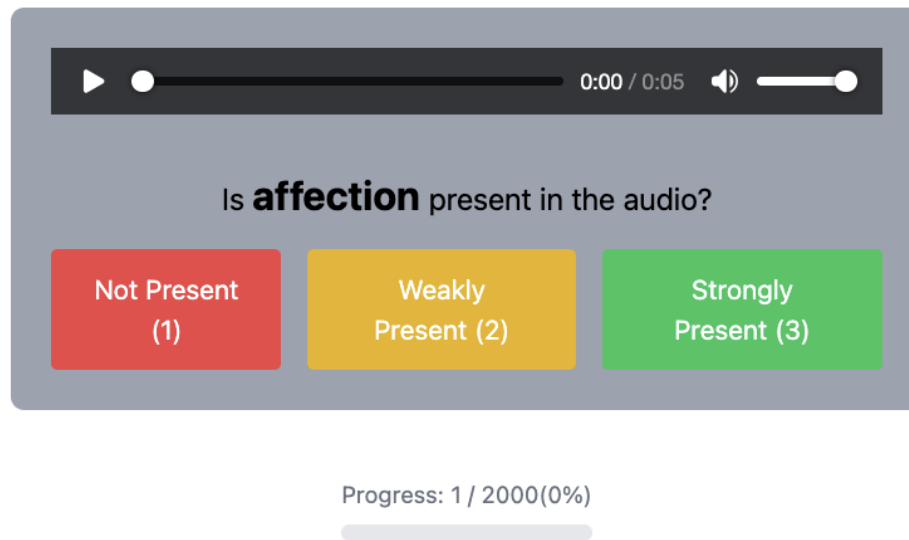


Figure 4: UI of our expert annotation tool for VOICENET-EMO.

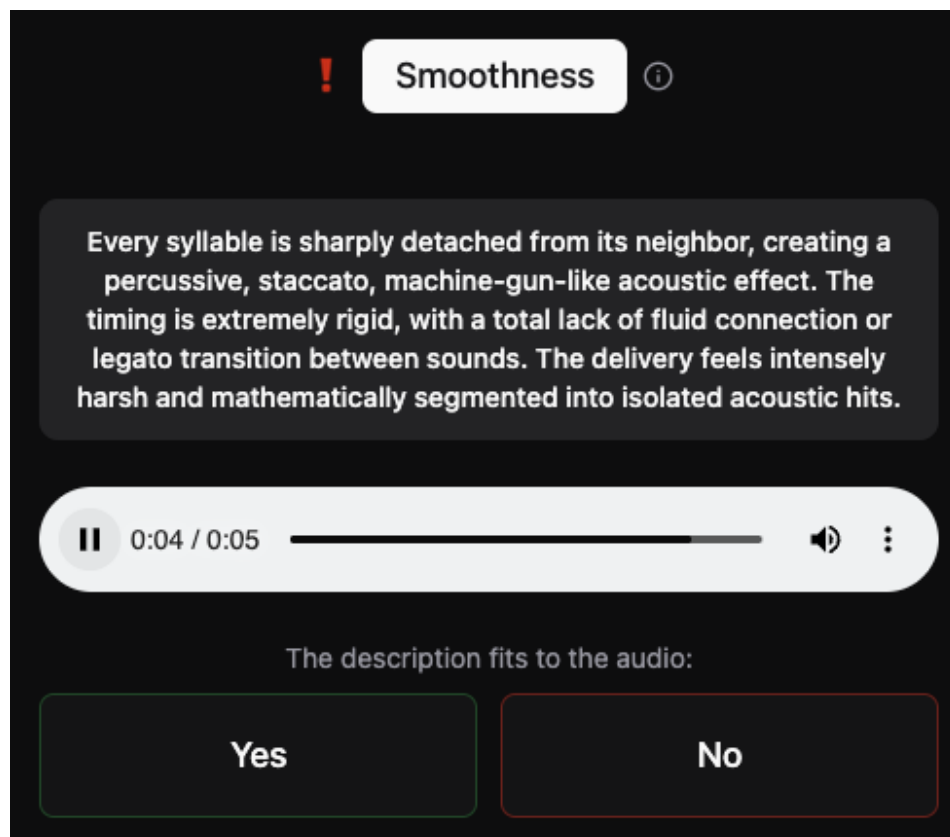


Figure 5: UI of the second annotation interface, used for the emotion-presence binary confirmation task on VOICENET-EMO. Annotators listen to a clip and confirm or reject whether the target emotion is present.

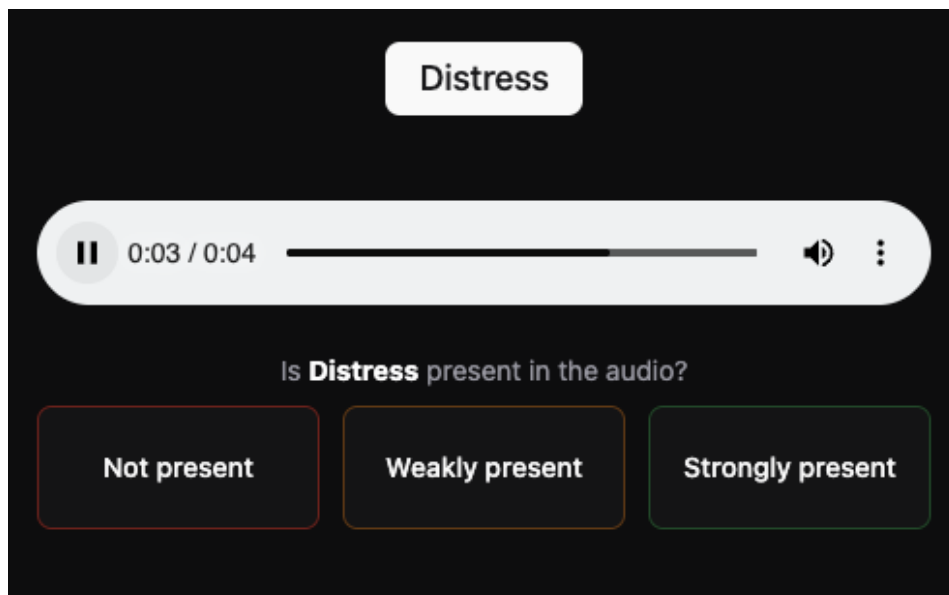


Figure 6: UI of the second annotation interface, used for the talking-style attribute binary confirmation task on VOICENET-EXT. For each (dimension, level) bucket, annotators confirm or reject whether the audio matches the target rubric-level description.